

MINIMAX REGRET TREATMENT RULES WITH FINITE SAMPLES WHEN A QUANTILE IS THE OBJECT OF INTEREST

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Consider a setup in which a decision maker is informed about the population by a finite sample and, based on that sample, has to decide whether or not to apply a certain treatment. We work out finite sample minimax regret treatment rules under various sampling schemes when potential outcomes are restricted onto the unit interval. In contrast to Stoye (2009, *Journal of Econometrics* 151, 70–81), where the focus is on the maximization of expected utility, the focus here is instead on a particular quantile of the outcome distribution. We find that in the case where the sample consists of a fixed number of untreated and a fixed number of treated units, any treatment rule is minimax regret optimal. The same is true in the case of random treatment assignment in the sample with any assignment probability and in the case of testing an innovation when the known quantile of the untreated population equals $1/2$. However, if the known quantile exceeds $1/2$, then never treating is the unique optimal rule, and if it is smaller than $1/2$, always treating is optimal. We also consider the case where a covariate is included.

1. INTRODUCTION

Consider a setup in which a decision maker (DM) has to decide whether or not to assign a treatment after observing data with observations from the untreated and

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treated populations.¹ The DM could be a policymaker who applies a treatment to the entire population or a person who is applying the treatment to an individual (e.g., herself). As examples of the latter setup, think of an individual who chooses between two hotels on an internet platform after observing a certain number of ratings or a medical doctor who considers a treatment for a patient after observing a certain number of outcomes from treated and untreated patients.

This article adds to a short but growing literature on finding treatment rules that have finite-sample optimality properties. In most of the literature, the focus is on the expected outcome under the chosen treatment rule.² Given that typically there is no treatment rule that is uniformly best over all possible joint distributions for (Y_0, Y_1) , where Y_0 and Y_1 denote random variables for outcomes without and with treatment, respectively, one has to resort to other criteria for optimality. One option is to consider a prior over the space of joint distributions and maximize expected outcome for this particular prior. Another option is to focus on admissible treatment rules, but that criterion typically does not single out an individual treatment rule (see Manski and Tetenov, 2023; Montiel Olea et al., 2026). Alternatively, one might consider finding a treatment rule that maximizes minimal expected outcome where the minimum is taken over all joint distributions of (Y_0, Y_1) . However, if there exists a distribution that assigns the minimal possible values in the shared domains of Y_0 and Y_1 with probability one, then any treatment rule is going to be optimal according to this criterion and therefore also the “max–min” approach is not pinning down a unique rule. For that reason, instead, the so-called “minimax regret” criterion is often adopted that determines treatment rules that minimize the maximal regret where regret is defined as the difference between the largest expected outcome that could be achieved for any treatment rule and the expected outcome under the treatment rule under consideration, and the maximum is taken with respect to all possible distributions for (Y_0, Y_1) . Stoye (2009) derives minimax regret rules in finite samples for the case of two treatments under various sampling schemes, namely, matched pairs, random sampling, and testing an innovation, and furthermore provides near-uniqueness results.

In this project, we are interested in a setup where, rather than expected outcome, the DM is concerned about the α -quantile of the outcome (for a given $\alpha \in [0, 1]$). Wang et al. (2018) consider robust estimation of the quantile-optimal

¹A treatment rule δ , formally introduced in (2.2) below, is a mapping from the data w_N (with N denoting the sample size) into the unit interval, with $\delta(w_N)$ denoting the probability with which treatment is assigned. Therefore, randomized treatment assignment is allowed for.

Alternatively, rather than framing the options as “treatment” and “no treatment,” one could frame the setup as a choice between two treatments.

²See, for example, Manski (2004), Manski and Tetenov (2007), Stoye (2007, 2009, 2012), Tetenov (2012), Masten (2023), Yata (2023), Chen and Guggenberger (2026), Kitagawa, Lee, and Qiu (2026), Montiel Olea, Qiu, and Stoye (2026), and additional references in these papers. Hirano and Porter (2009), Kitagawa and Tetenov (2018), Christensen, Moon, and Schorfheide (2026), and many other references therein also use the minimax regret criterion but consider an asymptotic, rather than a finite sample, framework. This literature is inspired by the classical work of Wald (1950). In a recent paper, Manski and Tetenov (2023) study several potential features of the state-dependent distribution of loss (rather than just its expectation) that a treatment rule generates across potential samples.

treatment regime and provide arguments as to why focusing on a quantile rather than the mean may be sensible in certain applications. In fact, in many applications, the tail of the outcome distribution is at the center of interest. For example, when evaluating job training programs to improve earnings, the focus is often on earnings in the lower tail, and likewise in survival analysis (e.g., survival time of cancer patients), the lower tail is of key importance. Wang et al. (2018, Sect. 2) provide numerical evidence where the mean-optimal treatment is detrimental for patients in the lower tail. In Economics, one might be interested in median income (case $\alpha = 1/2$) or a minimal level of educational achievement for schoolchildren, “no child left behind” (case $\alpha = 0$). See Manski (1988) who studies the “quantile utility” model whose predictions, unlike the expected utility model, are invariant under ordinal transformations of utility. Subsequently, Rostek (2010) axiomatizes quantile preference and, recently, Manski and Tetenov (2023) suggest considering various deviations from mean loss including quantiles. Also, see Chambers (2009).³

As the main contribution of this article, we derive minimax regret treatment rules δ in finite samples when an α -quantile of the outcome distribution is the focus of interest and outcomes Y_0 and Y_1 take values in the unit interval. That is, the objective function for the DM is $q_{s,\alpha}(Y_{B(\delta(w_N))})$, where $Y_{B(\delta(w_N))}$ denotes random outcomes generated when the treatment rule δ is used, with B representing a Bernoulli random variable that equals 1 with probability $\delta(w_N)$, and $q_{s,\alpha}(Y_{B(\delta(w_N))})$ denotes the α -quantile of $Y_{B(\delta(w_N))}$ when the joint distribution of (Y_0, Y_1) equals s .

Somewhat surprisingly, we show that under various sampling schemes *all* treatment rules are minimax regret, namely, in case i), when the sample consists of a fixed number of treated and a fixed number of untreated units (i.e., unbalanced panels are allowed for) and in case ii), under random assignment with arbitrary treatment assignment probability in the sample equal to $p \in (0, 1)$. In both cases i) and ii), maximal regret equals 1 for any treatment rule.

On the other hand, in case iii), “testing an innovation,” that is, the case where only data on Y_1 is observed and the α -quantile $q_{s,\alpha}(Y_0)$ of Y_0 is known, where again the subscript s denotes the joint distribution of (Y_0, Y_1) , if $q_{s,\alpha}(Y_0) > 1/2$, then $\delta \equiv 0$ is the unique minimax rule, if $q_{s,\alpha}(Y_0) < 1/2$, then $\delta \equiv 1$ is a minimax rule, and finally, if $q_{s,\alpha}(Y_0) = 1/2$, then any treatment rule δ is minimax regret; in each case, for minimax treatment rules δ , we obtain the formula $\min\{q_{s,\alpha}(Y_0), 1 - q_{s,\alpha}(Y_0)\}$ for the maximal regret.

Obviously, like for the case where expected outcomes are the focus, also here the max–min criterion is not informative. Namely, as long as the joint distribution of (Y_0, Y_1) can be chosen such that $q_{s,\alpha}(Y_0) = q_{s,\alpha}(Y_1) = 0$, that is, the worst possible outcome under the given restriction that $Y_0, Y_1 \in [0, 1]$, any treatment rule would

³Related (but in a non-finite-sample setup), Qi et al. (2019) and Qi, Pang, and Liu (2023) consider optimal decision rules based on the conditional value at risk (CVaR) measure. Quantile preferences have been attracting growing interest in the literature, for example, De Castro, Galvao, and Ota (2026) consider a model in which an economic agent maximizes the discounted value of a stream of future α -quantile utilities.

be max–min optimal. In contrast though, to the case where expected outcomes are the focus, for quantiles also minimax regret is not informative for sampling designs i) and ii) and also for iii) when $q_{s,\alpha}(Y_0) = 1/2$.

The typical strategy in the extant literature for determining minimax regret rules when the focus is on expected outcomes is via a Nash equilibrium approach in a fictitious zero sum game in which the DM plays against an antagonistic nature whose payoff equals regret. To establish that a particular treatment rule δ is minimax regret, one attempts to guess a “state of nature” s , allowed to be mixed strategy (over a finite number of states), called a *least favorable distribution*, for which the pair (δ, s) constitutes a Nash equilibrium. Existence of such a pair (δ, s) implies that δ is indeed a minimax regret rule (see, e.g., Berger, 1985; Stoye, 2009; Chen and Guggenberger, 2026). Often, in a first step, one restricts outcomes to be Bernoulli and finds a minimax rule in this simplified setup and then, in a second step, uses the so-called coarsening approach to tackle the general case (see, e.g., Cucconi, 1968; Gupta and Hande, 1992; Schlag, 2003, 2006). When nature picks a state of the world trying to inflict high regret, it faces the trade-off that, on the one hand, a high differential between expected outcomes with and without treatment is needed, but, on the other hand, the more different the distributions of Y_0 and Y_1 are the easier the DM can tell them apart using the sample information.

The proof structure for the main results in this article differs from the one just described. Namely, for cases i) and ii), we show that for any treatment rule δ , one can find a state of nature $s = s_\delta$ such that s inflicts the highest possible regret, namely, 1. That insight is sufficient to establish that all rules are minimax regret and that maximal regret equals 1 for all treatment rules. It is noteworthy and remarkable that, despite the trade-off just described, nature is powerful enough to inflict maximal regret on the DM. Even more surprisingly, the conclusions under i) and ii) continue to be true even if nature is restricted to only Bernoulli distributions. The proof for part iii) relies on the main insight that if $\delta \neq 0$, then there exists a state of nature such that regret equals the known α -quantile of Y_0 . For that to be true, it is sufficient for nature to have discrete distributions, supported on $N + 1$ points, at its disposal.

We start off with the case with no covariates and then allow for a discrete-valued covariate in the model. In the latter case, a treatment rule maps a sample onto treatment probabilities for each of the K possible outcomes of the covariate. Not surprisingly, given the results without a covariate, we find that in cases i) and ii), and case iii) with known quantile equal to 1/2, again, each treatment rule is minimax regret, while in case iii) when the known quantile is different from 1/2, no-data rules are minimax regret.

We include a small finite sample simulation study in the case of “testing an innovation” where we simulate regret of various treatment rules, namely, the empirical success rule and several no-data rules. The study corroborates our theoretical findings about minimax regret treatment rules and the formulas for maximal regret that we derive.

The remainder of the article is organized as follows. Section 2 introduces the theoretical setup for our notion of a minimax regret rule with quantiles and contains analytical results when no covariates are included. In Section 2.1, we present various approaches for how quantiles could be incorporated into the Waldian framework of statistical decision theory and juxtapose them with our approach. Section 2.2 derives minimax regret treatment rules under various sampling schemes and also provides a brief discussion of minimax regret treatment rules when certain restrictions are imposed on the states of nature. Section 2.3 simulates the performance of various treatment rules in finite samples in the case of testing an innovation. Finally, Section 3 derives minimax regret treatment rules in the case where covariates are included in the model. All proofs are given in the Appendix.

2. THEORETICAL SETUP WITHOUT COVARIATES

A DM has to decide whether or not to assign treatment after being informed about the population by a finite sample. The setup is very similar to the one in Stoye (2009) with one key modification. Namely, rather than focusing on mean outcomes, here we are concerned with the α -quantile of the outcome distribution.

For most parts of the article, potential outcomes Y_0 and Y_1 for untreated/treated individuals are restricted to the unit interval

$$Y_0, Y_1 \in S := [0, 1], \quad (2.1)$$

where S is assumed known to the DM. At first, different members of the population are all identical to the DM. Later, we will consider the case where a covariate is included and treatment assignment can be made conditional on the realization of the covariate.

By $\mathbb{S}$, we denote the set of “states of the world” s , where an s denotes a possible joint probability distribution over the potential outcomes for Y_0 and Y_1 . Unless otherwise stated, $\mathbb{S}$ is unrestricted and contains all possible joint distributions for Y_0 and Y_1 on S^2 . Upon observing a sample $w_N = (t, y)$ of size N of treatment statuses $t = (t_1, \dots, t_N)$ and outcomes $y = (y_1, \dots, y_N)$, where the i -th component of y , y_i , is an independent realization of Y_{t_i} and t_i denotes the treatment received by individual i , the task for the DM is to choose

$$\delta(w_N) \in [0, 1], \quad (2.2)$$

which denotes the probability with which treatment is assigned.⁴ Namely, which treatment the DM assigns is determined as an independent draw of the Bernoulli random variable $B = B(\delta(w_N)) \in \{0, 1\}$ that equals 1 with probability $\delta(w_N)$ and is assumed to be independent of all other random objects. Therefore, the setup here allows for randomized treatment rules.

⁴We do not index t and y by N because it would make the notation too cumbersome.

Let $\alpha \in [0, 1]$. Given a state of the world s and a statistical treatment rule δ , the objective function for the DM is

$$u(\delta, s) = q_{s, \alpha}(Y_{B(\delta(w_N))}), \quad (2.3)$$

where $Y_{B(\delta(w_N))}$ denotes random outcomes generated when the treatment rule δ is used, and $q_{s, \alpha}(Y_{B(\delta(w_N))})$ denotes the α -quantile of $Y_{B(\delta(w_N))}$ when the state of the world is $s \in \mathbb{S}$. In particular, when the treatment rule $\delta(w_N)$ equals 0 (or 1), then with probability 1 $Y_{B(\delta(w_N))}$ equals Y_0 (or Y_1).

By definition, an α -quantile of a scalar-valued random variable X with domain D is any number $q \in D$ that satisfies

$$P(X \leq q) \geq \alpha \text{ and } P(X \geq q) \geq 1 - \alpha. \quad (2.4)$$

Clearly then, any α -quantile of Y_0 , Y_1 , and $Y_{B(\delta(w_N))}$ is an element of $[0, 1]$. In general, this definition does not lead to a unique α -quantile. The definition allows for the case where a quantile has a nonzero point mass. To be explicit in cases where there is non-uniqueness, we make the following definition that hinges on a choice $r \in [0, 1]$.

Definition (α -quantile): Let Q denote the set of all α -quantiles q . For $\alpha \in (0, 1)$, we define the α -quantile as

$$r \sup Q + (1 - r) \inf Q. \quad (2.5)$$

When $\alpha = 0$, we use $\sup Q$ (i.e., we use $r = 1$) and when $\alpha = 1$, we use $\inf Q$ (i.e., we use $r = 0$).

From now on, by $q_{s, \alpha}(Y_{B(\delta(w_N))})$, we denote the α -quantile of $Y_{B(\delta(w_N))}$ when the state of the world is s . For simplicity of notation, unless needed for clarity, we do not index that expression by r .

By definition, regret of the treatment rule δ for a given distribution s of (Y_0, Y_1) equals

$$R(\delta, s) = \sup_{d \in \mathbb{D}} u(d, s) - u(\delta, s), \quad (2.6)$$

where $\mathbb{D}$ denotes the set of all possible treatment rules. In words, regret equals the difference between the highest α -quantile that could have been achieved for any treatment rule $d \in \mathbb{D}$ and the α -quantile obtained for the particular treatment rule δ for the given state of nature s . If $q_{s, \alpha}(Y_1) \geq q_{s, \alpha}(Y_0)$ ($q_{s, \alpha}(Y_1) < q_{s, \alpha}(Y_0)$) and $d^* = 1$ ($d^* = 0$) is an element of $\mathbb{D}$, then $\sup_{d \in \mathbb{D}} u(d, s)$ is taken on by d^* and equals $q_{s, \alpha}(Y_1)$ ($q_{s, \alpha}(Y_0)$) (see Lemma 3(ii) in the Appendix). If restrictions are imposed on $\mathbb{D}$, then $\sup_{d \in \mathbb{D}} u(d, s)$ may not be taken on by any element in $\mathbb{D}$. Again, without restrictions, $1(q_{s, \alpha}(Y_1) \geq q_{s, \alpha}(Y_0))$ is an infeasible optimal rule, where $1(\cdot)$ denotes the indicator function.

In this article, we focus on minimax regret treatment rules. By definition, if it exists, such a rule satisfies

$$\delta^* \in \arg \min_{\delta \in \mathbb{D}} \sup_{s \in \mathbb{S}} R(\delta, s). \quad (2.7)$$

In contrast, a maximin treatment rule, if it exists, satisfies

$$\delta^+ \in \arg \max_{\delta \in \mathbb{D}} \inf_{s \in \mathbb{S}} u(\delta, s). \quad (2.8)$$

As discussed already elsewhere (see, e.g., Manski, 2004; Stoye, 2009), the maximin criterion may lead to the uninformative result that *all* $\delta \in \mathbb{D}$ are maximizers. That occurs, for example, if for a particular $s^+ \in \mathbb{S}$, $u(\delta, s^+)$ does not depend on δ and takes on its smallest possible value, $u(\delta, s^+) = \inf_{s \in \mathbb{S}} u(\delta, s)$. That situation also occurs in our setup where the objective is to maximize the quantile of the outcome distribution, namely, when $s^+ \in \mathbb{S}$ is chosen such that $q_{s, \alpha}(Y_0) = q_{s, \alpha}(Y_1) = 0$. In contrast to the setup where expected outcomes are the objective, as we will establish next, it turns out that for quantiles, depending on the particular sampling design, also the minimax regret criterion may be uninformative.

2.1. Waldean Statistical Decision Theory and Quantiles⁵

In this section, we further discuss the proposed criterion introduced in (2.3) and juxtapose it with other possible approaches of how quantiles could be incorporated into a Waldean framework of statistical decision theory.

In a series of pathbreaking contributions, Wald (1945, 1947, 1950) introduced a framework for statistical decision theory whose main components are a statistical model, an action space, statistical decision rules, a welfare and an expected welfare function, and finally an evaluation criterion. Instead of welfare and expected welfare the presentation could be framed in terms of loss (defined as negative welfare) and risk (defined as expected loss). See Hirano (2026) for a comprehensive review that also includes contemporary developments.

In the particular context considered here where the DM needs to pick treatment 0 or 1 after observing the i.i.d. sample w_N , a minimax regret rule in the Waldean formulation solves

$$\arg \min_{\delta \in \mathbb{D}} \max_{s \in \mathbb{S}} [\max\{\mu_0, \mu_1\} - E_s Y_{B(\delta(w_N))}], \quad (2.9)$$

where $\mu_t = E_s Y_t$ for $t = 0, 1$ and E_s denotes the expectation operator under s . It is easily shown that s matters only via (μ_0, μ_1) . By the law of iterated expectations,

$$\begin{aligned} E_s Y_{B(\delta(w_N))} &= E_{w_N} E_s (Y_{B(\delta(w_N))} | w_N) \\ &= \mu_0 E_{w_N} (1 - \delta(w_N)) + \mu_1 E_{w_N} \delta(w_N), \end{aligned} \quad (2.10)$$

overall uncertainty can be separated into *sampling uncertainty* through w_N , (potential) randomness through the *treatment assignment* B , and randomness through the *potential outcome variables* (Y_0, Y_1) ; *risk* is defined as the average loss over sampling uncertainty.

To adapt the Waldean framework to one that is based on the notion of α -quantile rather than expectation, we propose replacing expectations by α -quantiles in the

⁵This section is inspired by the constructive comments of an anonymous referee.

formulation (2.9), that is, we suggest solving (2.7) which is

$$\arg \min_{\delta \in \mathbb{D}} \max_{s \in \mathbb{S}} [\max\{q_{s,\alpha}(Y_0), q_{s,\alpha}(Y_1)\} - q_{s,\alpha}(Y_{B(\delta(w_N))})]. \quad (2.11)$$

Given that there is no equivalent to the law of iterated expectations for quantiles, by doing so, one loses the separation of the regret criterion into loss without sampling uncertainty and sampling uncertainty. Our proposed criterion aggregates joint uncertainty (sampling, treatment assignment, and outcome) before taking the quantile.⁶

Instead, to maintain such separation, one could first define regret loss given a realized sample w_N as

$$L(\delta(w_N), s) = \max\{q_{s,\alpha}(Y_0), q_{s,\alpha}(Y_1)\} - q_{s,\alpha}(Y_{B(\delta(w_N))} | w_N) \quad (2.12)$$

and then define an alternative regret to (2.6) as the average of regret loss over the sampling distribution w_N

$$R'(\delta, s) = \max\{q_{s,\alpha}(Y_0), q_{s,\alpha}(Y_1)\} - E_{w_N} q_{s,\alpha}(Y_{B(\delta(w_N))} | w_N). \quad (2.13)$$

The first term in (2.12) is a benchmark attainable if one could pick the better treatment in terms of quantile outcome without sampling uncertainty; the second term is the α -quantile of $Y_{B(\delta(w_N))}$ conditional on the realized sample w_N . This criterion may be misaligned when the DM cares about tails in the distribution of $q_{s,\alpha}(Y_{B(\delta(w_N))} | w_N)$ over realized samples rather than the average over sampling uncertainty. As yet another alternative, one could consider a regret function defined as the β -quantile (for some β in the unit interval) of the loss function in (2.12), that is,

$$R''(\delta, s) = q_{s,\beta}(L(\delta(w_N), s)). \quad (2.14)$$

By the monotone transform identity of quantiles, it follows that

$$R''(\delta, s) = \max\{q_{s,\alpha}(Y_0), q_{s,\alpha}(Y_1)\} - q_{s,1-\beta}(q_{s,\alpha}(Y_{B(\delta(w_N))} | w_N)) \quad (2.15)$$

and the interpretation of the criterion is not straightforward.

Manski and Tetenov (2023, Sect. 6.1) suggest an entire class of alternative approaches that also maintain separation. Starting with an arbitrary loss function (for instance, negative welfare) for a given action by the DM (i.e., in our setup, a choice of treatment 0 or 1) and DGP s , the suggestion is to consider, for example, an α -quantile of the loss function over the sampling distribution. In our setup, such an approach could be formulated as

$$\arg \min_{\delta \in \mathbb{D}} \sup_{s \in \mathbb{S}} [\max\{\mu_0, \mu_1\} - q_{s,\alpha}(\mu_{B(\delta(w_N))})]. \quad (2.16)$$

⁶Note that $q_{s,\alpha}(Y_{B(\delta(w_N))} | w_N)$ does not in general equal $\delta(w_N)q_{s,\alpha}(Y_1) + (1 - \delta(w_N))q_{s,\alpha}(Y_0)$ in cases where $\delta(w_N) \in (0, 1)$.

One issue with the implementation of the various criteria is that analytical formulas are not generally available.⁷

We do not have a strong opinion about which one of the above criteria is generally preferable. However, we find our criterion very natural in situations where a DM applies treatment to an individual (e.g., herself) after observing the sample. Consider, for example, an individual who books a hotel accommodation on an internet platform, picking one of two options after observing a certain number of ratings on each (or picks one of two restaurants after having observed a number of ratings for each on the internet). The outcome is the rating the DM assigns to the hotel she booked, which could be interpreted as a proxy for the welfare the DM received from staying at the hotel. Another example is a medical doctor who picks one of two treatments for a patient after observing a certain number of outcomes on the two treatments.

In that type of example, when repeating the exercise, every observed outcome combines the randomness of the sample, (potential) randomness of treatment assignment, and the randomness of (Y_0, Y_1) . If the DM is concerned about quantiles of the outcome distribution, it seems that the criterion proposed in (2.11) is a natural choice.

2.2. Minimax Regret Treatment Rules

Recall that a sample of size N has the general structure $w_N = (t, y)$. We consider three different sample designs, namely:

(i) *Fixed number of untreated/treated units* with $N_0 \in \{0, \dots, N\}$ i.i.d. observations of Y_0 and $N_1 := N - N_0$ i.i.d. observations of Y_1 for some $N \in \mathbb{N} \cup \{0\}$. Here, the notation for a sample can be simplified by dropping t . We write $w_N = (y(0)', y(1)')' \in [0, 1]^N$, where $y(0) \in [0, 1]^{N_0}$ contains the N_0 observations on untreated units and $y(1) \in [0, 1]^{N_1}$ contains the N_1 observations on the treated units. A treatment rule $\delta \in \mathbb{D}$ is then any mapping $[0, 1]^N \rightarrow [0, 1]$. It assigns a treatment probability $\delta(w_N) \in [0, 1]$ after observing the sample.

(ii) *Random assignment* with $N \in \mathbb{N} \cup \{0\}$ i.i.d. observations, where in the sample, the treatment probability equals $p \in (0, 1)$.⁸ A treatment rule $\delta \in \mathbb{D}$ is any mapping $\{0, 1\}^N \times [0, 1]^N \rightarrow [0, 1]$. The rule δ assigns a treatment probability $\delta(w_N) \in [0, 1]$ after observing a sample $w_N = (t, y)$ of treatment statuses $t = (t_1, \dots, t_N)$ and realizations $y = (y_1, \dots, y_N)$, where the i -th component of y , y_i , is an independent realization of Y_{t_i} .

(iii) *Testing an innovation* is the case where aspects of the distribution of Y_0 are known to the DM; in particular, we assume that the α -quantile of Y_0 is known (but nature can pick arbitrary distributions for Y_0 subject to that restriction). That is, in this case, the set $\mathbb{S}$ consists of all joint distributions s for (Y_0, Y_1) with the restriction

⁷Recently suggested numerical procedures for the implementation of minimax rules by Aradillas Fernández et al. (2025) and Guggenberger and Huang (2025) might be applicable also for these scenarios.

⁸Note that $p \in \{0, 1\}$ leads back to design (i) with N_0 or N_1 equal to N .

that the α -quantile of the marginal for Y_0 equals a certain value $q_\alpha(Y_0)$. In this case, $N \in \mathbb{N} \cup \{0\}$ i.i.d. observations of Y_1 are observed. Here again, the notation for a sample can be simplified. We write $w_N = y(1) \in [0, 1]^N$, where $y(1)$ contains the N observations of the treated units. A treatment rule $\delta \in \mathbb{D}$ is any mapping $[0, 1]^N \rightarrow [0, 1]$. The rule δ assigns a treatment probability $\delta(w_N) \in [0, 1]$ after observing a sample $w_N = (y_{1,1}, \dots, y_{1,N})$ of N independent realizations $y_{1,i}$, $i = 1, \dots, N$ of Y_1 .

The designs above nest the ones in Stoye (2009). In contrast to Stoye (2009), in design (i), we allow for arbitrary numbers of treated and untreated units rather than $N/2$ units each as in “matched pairs” and in design (ii), the treatment probability is any fixed number $p \in (0, 1)$ rather than necessarily $p = 0.5$.

The following statement provides the analog to Proposition 1 in Stoye (2009) and is the main contribution of this article.

PROPOSITION 1. (i) *Consider the case where the sample consists of a “fixed number of treated/untreated units.” If $(\alpha \in (0, 1)$ and $r \in [0, 1])$ or $(\alpha = 0$ and $r = 1)$, then any treatment rule $\delta \in \mathbb{D}$ is minimax regret and $\max_{s \in \mathbb{S}} R(\delta, s) = 1$ for any $\delta \in \mathbb{D}$. If $(\alpha = 1$ and $r = 0)$, then exactly those treatment rules $\delta \in \mathbb{D}$, that are not equal to 1 wpl and not equal to 0 wpl, are minimax regret and satisfy $\max_{s \in \mathbb{S}} R(\delta, s) = 0$.*

(ii) *Consider the case where the sample is generated via “random assignment.” Then the same statement as in part (i) holds.*

(iii) *In the case of “testing an innovation,” let $(\alpha \in (0, 1)$ and $r \in [0, 1])$ or $(\alpha = 0$ and $r = 1)$. If the known α -quantile of Y_0 , $q_\alpha(Y_0)$, equals $1/2$, then any treatment rule $\delta \in \mathbb{D}$ is minimax regret; if instead $q_\alpha(Y_0) > 1/2$, then $\delta^0 \equiv 0$ is the unique minimax regret rule; if $q_\alpha(Y_0) < 1/2$, then $\delta^1 \equiv 1$ is a minimax regret rule; in each case, for minimax regret treatment rules δ , we obtain $\max_{s \in \mathbb{S}} R(\delta, s) = \min\{q_\alpha(Y_0), 1 - q_\alpha(Y_0)\}$. If $(\alpha = 1$ and $r = 0)$, then exactly those treatment rules $\delta \in \mathbb{D}$, that are not equal to 1 wpl and not equal to 0 wpl, are minimax regret and satisfy $\max_{s \in \mathbb{S}} R(\delta, s) = 0$.*

Comments. 1. Proposition 1 establishes that the minimax regret criterion when applied to α -quantiles does not favor data-driven rules. In fact, in the “testing an innovation” case, data-driven rules are strictly dominated by $\delta^0 \equiv 0$ when $q_\alpha(Y_0) > 1/2$ and weakly dominated by $\delta^1 \equiv 1$ when $q_\alpha(Y_0) < 1/2$. When $q_\alpha(Y_0) = 1/2$, all treatment rules are minimax. This is in stark contrast to the results in Stoye (2009) about minimax regret treatment rules when the focus is on mean outcomes. Namely, Stoye (2009) shows that, for example, in the case of binary outcomes, where the sample is obtained as a matched pair, to be minimax regret optimal, the treatment that has more successes in the sample must be chosen with probability one.

2. To provide intuition of the result in (i) and (ii), assume $\alpha \in (0, 1)$ and consider first the simplest case where the sample size N is 0. In that case, a treatment rule δ is simply an element in $[0, 1]$ that denotes the probability of assigning treatment. For given δ , the objective for nature is to find a distribution for (Y_0, Y_1) such that

the α -quantiles of the marginals have maximal distance (i.e., distance 1) and such that the α -quantile of $Y_{B(\delta)}$ is zero. Let $\delta > 0$ first. Assume nature picks Y_0 and Y_1 as independent Bernoulli

$$P(Y_1 = 0) = 1, P(Y_0 = 0) = \alpha - \varepsilon \text{ for some } \varepsilon \in (0, \alpha] \quad (2.17)$$

and, consequently, $P(Y_0 = 1) = 1 - (\alpha - \varepsilon)$. Thus, $q_{s,\alpha}(Y_1) = 0$ and $q_{s,\alpha}(Y_0) = 1$ and

$$P(Y_{B(\delta)} = 0) = \delta + (1 - \delta)(\alpha - \varepsilon). \quad (2.18)$$

Thus, $P(Y_{B(\delta)} = 0) > \alpha$ iff $\delta(1 - \alpha) > \varepsilon(1 - \delta)$ which holds for ε small enough. Thus,

$$\max_{s \in \mathbb{S}} R(\delta, s) = 1 \quad (2.19)$$

for all $\delta \in (0, 1]$. If instead $\delta = 0$, then nature chooses $P(Y_1 = 1) = 1$ and $P(Y_0 = 0) = 1$ which leads to regret of 1.

This result may be surprising at first. For example, if the DM tries to be completely balanced and picks $\delta = 1/2$ (which seems reasonable in the no data case), nature can inflict regret equal to one by picking $P(Y_1 = 0) = 1$ and, for example, $P(Y_0 = 0) = \alpha - \min\{\alpha/2, (1 - \alpha)/2\}$. This is in stark contrast to the case considered in Stoye (2009) where the DM cares about expected outcome under δ , that is, $\mu_0(1 - \delta) + \mu_1\delta$, where μ_t for $t = 0, 1$ denotes the expectation of Y_t under s . Regret for given δ and s (which only matters via (μ_0, μ_1)) then equals $\max\{\mu_0, \mu_1\} - (\mu_0(1 - \delta) + \mu_1\delta)$. When the DM picks a δ with $\delta \geq 1/2$, then the maximal regret nature can inflict equals δ obtained for any s with $E_s Y_0 = 1$ and $E_s Y_1 = 0$. Therefore, the DM's minimax regret choice is $\delta = 1/2$. What explains the different results with quantiles and expectations? What drives the results with quantiles is the discontinuity of the α -quantile of a random variable X with respect to the cdf F_X of X . In the above construction, the random variables Y_0 and $Y_{B(\delta)}$ have cdfs that are uniformly "very close" yet their α -quantiles differ by 1. For example, take again $\delta = 1/2$ and consider $\alpha = 0.99$. Then both Y_0 and $Y_{B(\delta)}$ are Bernoulli with $P(Y_0 = 0) = 0.99 - 0.005 = 0.985$ and $P(Y_{B(\delta)} = 0) = 1/2 + 1/2 \cdot 0.985 = 0.9925$. Thus, their cdfs are almost identical but their 0.99-quantiles differ by 1. Instead the expectations of these two random variables are very close.

Surprisingly, the intuition of the no-data example generalizes to cases with arbitrary sample size $N > 0$. Again, for any given treatment rule $\delta \in \mathbb{D}$, one constructs an $s_\delta \in \mathbb{S}$ for which $\max\{q_{s_\delta, \alpha}(Y_0), q_{s_\delta, \alpha}(Y_1)\} = 1$ and $u(\delta, s_\delta) = 0$ and thus $R(\delta, s_\delta) = 1$. Again $s_\delta \in \mathbb{S}$ can be chosen such that Y_0 and Y_1 are independent and both have Bernoulli distributions. That is, s_δ is then fully described by the two parameters $P_{s_\delta}(Y_0 = 0)$ and $P_{s_\delta}(Y_1 = 0)$. Denote by 0^{N_0} an N_0 -dimensional column vector of zeros.

In case (i) of Proposition 1, if δ is such that $\delta((0^{N_0}, v')) = 1$ for all $v \in \{0, 1\}^{N_1}$, then nature attempts punishing the DM for always using treatment 1 when no successes are observed for treatment 0, by choosing $P_{s_\delta}(Y_1 = 0) = 1$ and by choosing $P_{s_\delta}(Y_0 = 0)$ "big enough" that only zeros are observed for the untreated

individuals “sufficiently often” guaranteeing $Y_{B(\delta(w))}$ has α -quantile 0, but “small enough” that Y_0 has α -quantile 1. The proof shows that this can indeed be achieved by picking $P_{s_\delta}(Y_0 = 0) = \alpha - \varepsilon$ for some small enough $\varepsilon > 0$.

If on the other hand δ is such that $\delta((0^{N_0'}, v')') < 1$ for at least one $v \in \{0, 1\}^{N_1}$, then nature attempts punishing the DM for not using treatment 1 often enough when no successes are observed for treatment 0, by choosing $P_{s_\delta}(Y_0 = 0) = 1$ and by choosing $P_{s_\delta}(Y_1 = 0)$ “small enough” that Y_1 has α -quantile 1 but “big enough” that $Y_{B(\delta(w))}$ has α -quantile 0. The proof shows that this can indeed be achieved by picking $P_{s_\delta}(Y_1 = 0) = \alpha - \varepsilon$ for some small enough $\varepsilon > 0$.

A surprising fact about the proof is that exploiting properties of δ on the set of samples $\{(0^{N_0'}, v')', v \in \{0, 1\}^{N_1}\}$ alone gives nature enough leverage to inflict maximal regret.

Next, we provide additional insights for this fact. The proof of Proposition 1(i) reveals that for any $\delta \in \mathbb{D}$, one can find $s_\delta \in \mathbb{S}$ with Y_0 and Y_1 independent Bernoulli distributions such that $R(\delta, s_\delta) = 1$. In fact, the proof reveals that for any treatment rule δ for which

$$\delta^{0,j} := \sum_{v \in \{0, 1\}^{N_1}, \bar{v} = j} \delta((0^{N_0'}, v')') < \binom{N_1}{j} \quad (2.20)$$

for at least one $j = 0, \dots, N_1$ (where $\bar{v}$ denotes the number of observations in the sample for which Y_1 equals 1), one can achieve $R(\delta, s_\delta) = 1$ when Y_0 and Y_1 are independent Bernoulli distributions with $Y_0 = 0$ with probability 1. For such a treatment rule δ , we now discuss regret of δ as a function of $a \in [0, 1]$ when nature picks $s_a \in \mathbb{S}$ and under s_a , $Y_0 = 0$ with probability 1 and Y_1 is Bernoulli with $P_{s_a}(Y_1 = 0) = a$. We will focus on the case $\alpha = 1/2$, $r = 0$, and on two particular treatment rules, namely, $\delta^{0.5} \equiv 1/2$ (the treatment rule that always picks $1/2$ independent of the data w_N) and the empirical success rule δ^{ES} , defined by

$$\delta^{ES}(w_N) = I(q_\alpha(Y_1, w_N) > q_\alpha(Y_0, w_N)) + .5I(q_\alpha(Y_1, w_N) = q_\alpha(Y_0, w_N)), \quad (2.21)$$

where $q_\alpha(Y_j, w_N)$ denotes the α -sample quantile of Y_j for the sample w_N for $j = 0, 1$ and the α -sample quantile is defined with $r = 0$ in (2.5). We will take $N_0 > 0$ and N_1 even. When $Y_0 = 0$ with probability 1, it follows that $q_\alpha(Y_0, w_N) = 0$ with probability 1 and $q_\alpha(Y_1, w_N) > 0$ occurs if strictly more than $N_1/2$ observations on Y_1 in the sample equal 1. (In that case, $q_\alpha(Y_1, w_N) = 1$; otherwise, $q_\alpha(Y_1, w_N) = 0$.) Note that $\delta^{0,j}$ in (2.20) equals $0.5 \binom{N_1}{j}$ for $\delta = \delta^{0.5}$; it equals $0.5 \binom{N_1}{j}$ if $j \leq N_1/2$ and $\binom{N_1}{j}$ if $j > N_1/2$ for $\delta = \delta^{ES}$.

From simple calculations in (A.6) in the Appendix, it follows that, for any $\delta \in \mathbb{D}$,

$$P_{s_a}(Y_{B(\delta(w_N))} = 0) = \sum_{j=0}^{N_1} [a^{N_1-j} (1-a)^j (a-1)] \delta^{0,j} + 1. \quad (2.22)$$

Note that $q_{s_a, \alpha}(Y_{B(\delta(w_N))}) = 0$ if $P_{s_a}(Y_{B(\delta(w_N))} = 0) \geq 0.5$ and $q_{s_a, \alpha}(Y_{B(\delta(w_N))}) = 1$ otherwise. It follows that regret equals 1 if $a < 1/2$ and $P_{s_a}(Y_{B(\delta(w_N))} = 0) \geq 0.5$ and regret equals 0 otherwise.

For $\delta = \delta^{0.5}$, it follows that $P_{sa}(Y_{B(\delta(w_N))} = 0) = 0.5a + 0.5$. Therefore, regret equals 1 if $a < 1/2$ and 0 otherwise. For $\delta = \delta^{ES}$, we can evaluate $P_{sa}(Y_{B(\delta(w_N))} = 0)$ given the formulas above. The results do not depend on $N_0 > 0$. We find that for a given N_1 regret equals 1 if $a \in [l(N_1), 0.5)$ and 0 otherwise, for an $l(N_1) < 0.5$. For example, if $N_1 = 10, 20, 30, 40$, and 50 , $l(N_1) = 0.394, 0.416, 0.426, 0.433$, and 0.437 , respectively.

3. In part (iii) of Proposition 1, in the case $q_\alpha(Y_0) < 1/2$, we could not rule out that there are other minimax regret rules besides $\delta^1 \equiv 1$ when $N > 0$. When $N = 0$, it is obvious that δ^1 is the unique minimax rule. A minimax regret treatment rule, if it is not unique, may be inadmissible. In the case of “testing an innovation,” $\delta^0 \equiv 0$ (when $q_\alpha(Y_0) > 1/2$) is admissible, but we have not determined whether that is true for δ^1 (when $q_\alpha(Y_0) < 1/2$) and for $\delta^{0.5} \equiv 1/2$ (when $q_\alpha(Y_0) = 1/2$).

4. In case $Y_0, Y_1 \in (0, 1)$ rather than $Y_0, Y_1 \in [0, 1]$, $\max_{s \in \mathbb{S}} R(\delta, s)$ does not generally exist when $\mathbb{S}$ denotes the set of all joint probability distributions over the potential outcomes for $Y_0, Y_1 \in (0, 1)$. For example, in cases (i) and (ii), nature choosing only distributions with support on $[\varepsilon, 1 - \varepsilon]$ for some small $\varepsilon > 0$, it can generate regret of $1 - 2\varepsilon$ and therefore $\sup_{s \in \mathbb{S}} R(\delta, s) = 1$ for the unrestricted space of distributions for $Y_0, Y_1 \in (0, 1)$. This result can be proven using the exact same proof technique as for Proposition 1. Therefore, if minimax rules are defined with $\sup_{s \in \mathbb{S}} R(\delta, s)$ (as we do in (2.7)) rather than $\max_{s \in \mathbb{S}} R(\delta, s)$, the results in Proposition 1(i) and (ii) continue to hold.

5. If $Y_0, Y_1 \in [0, \infty)$ rather than $Y_0, Y_1 \in [0, 1]$, then $\max_{s \in \mathbb{S}} R(\delta, s)$ does not typically exist when $\mathbb{S}$ denotes the set of all joint probability distributions over the potential outcomes for $Y_0, Y_1 \in [0, \infty)$. For example, in cases (i) and (ii), nature choosing only distributions with support on $[0, M]$ for $M > 0$, it can generate regret of M and therefore $\sup_{s \in \mathbb{S}} R(\delta, s) = \infty$ when $\mathbb{S}$ denotes the set of all joint distributions for $Y_0, Y_1 \in [0, \infty)$. This result can be proven using the exact same proof technique as for Proposition 1. Therefore, if minimax rules are defined with $\sup_{s \in \mathbb{S}} R(\delta, s)$ (as we do in (2.7)) rather than $\max_{s \in \mathbb{S}} R(\delta, s)$, then for $\alpha \in (0, 1)$, any treatment rule is minimax regret optimal in cases (i) and (ii).

Given the limit experiment results in Hirano and Porter (2009), it would also be interesting to study the case where potential outcomes are restricted to be normally distributed. However, in that case, one would need to employ other proof techniques than the ones used in the current article.

Restrictions on nature’s action space: In what follows, we impose various restrictions on nature’s action space and study the implications on the results obtained in Proposition 1. For simplicity, assume $\alpha \in (0, 1)$.

COROLLARY 1. (a) *The results in Proposition 1 (i) and (ii) remain valid when rather than $Y_0, Y_1 \in [0, 1]$, the setup is altered to $Y_0, Y_1 \in \{0, 1\}$. Similarly, the result in (iii) remains valid if the distributions of Y_0 and Y_1 are restricted to be discrete and supported on at most $N + 1$ points in $[0, 1]$.*

(b) *If $\mathbb{S}$ equals the set of distributions for $Y_0, Y_1 \in [0, 1]$ whose marginals are all continuous (with respect to Lebesgue measure), then Proposition 1 continues to hold.*

Comments. 1. Corollary 1(a) is a direct corollary from the proof of Proposition 1. In the proof of Proposition 1(i) and (ii), only Bernoulli distributions are used for nature, while in part (iii), only discrete distributions are used.

2. Corollary 1(b) considers the case where nature is restricted to continuous distributions. We have seen in Proposition 1 that in cases (i) and (ii) any treatment rule $\delta \in \mathbb{D}$ is minimax and $\max_{s \in \mathbb{S}} R(\delta, s) = 1$. Because without point masses, it is impossible for a random variable to have α -quantile equal to 0, it follows that $R(\delta, s)$ is always strictly smaller than 1 for any pair (δ, s) when s is continuous with respect to Lebesgue measure. The main construction in the proof of Proposition 1 still goes through when one considers a sequence of continuous distributions that converge to the Bernoulli distributions that are used in that proof. As a technical detail, it is important to use $\sup_{s \in \mathbb{S}} R(\delta, s)$ rather than $\max_{s \in \mathbb{S}} R(\delta, s)$ in the definition of regret, because $\max_{s \in \mathbb{S}} R(\delta, s)$ would not exist in the case considered here. One can establish that $\sup_{s \in \mathbb{S}} R(\delta, s) = 1$ for any treatment rule $\delta \in \mathbb{D}$.

3. In the case of “testing an innovation,” a restriction on nature’s action space occurs if one assumes that the entire distribution of Y_0 is known, not just its α -quantile. Namely, in that case, the set $\mathbb{S}$ consists of all possible distributions s for $Y_1 \in [0, 1]$ (while the distribution of Y_0 is given). The analysis of that problem is more difficult. Denote by F_{Y_0} and $q_\alpha(Y_0)$ the cdf and α -quantile of Y_0 , respectively.

Take $r = 0$ in (2.5) and assume $F_{Y_0}(q_\alpha(Y_0)) > \alpha$. Consider the case $N = 0$ in which case a treatment rule $\delta \in [0, 1]$ denotes the treatment probability. Under these assumptions, the following statements hold.

Denote by $q(\delta)$ the smallest $q \in [0, q_\alpha(Y_0)]$ such that

$$\delta + (1 - \delta)F_{Y_0}(q) \geq \alpha. \quad (2.23)$$

Then, when $q_\alpha(Y_0) < 1/2$, $\delta^1 \equiv 1$ is the only minimax regret rule and $\max_{s \in \mathbb{S}} R(\delta^1, s) = q_\alpha(Y_0)$. When $q_\alpha(Y_0) = 1/2$, any $\delta \in [0, 1]$ is minimax regret with maximal regret equal to $q_\alpha(Y_0)$. Finally, when $q_\alpha(Y_0) > 1/2$, $\delta \in [0, 1]$ is minimax regret iff

$$q_\alpha(Y_0) - q(\delta) \leq 1 - q_\alpha(Y_0). \quad (2.24)$$

Given that $q_\alpha(Y_0) - q(\delta)$ is a weakly increasing function in δ , if we denote by $\delta^* \in [0, 1]$ an intersection of $q_\alpha(Y_0) - q(\delta)$ and $1 - q_\alpha(Y_0)$, then any $\delta \in [0, \delta^*]$ is minimax regret and for those δ we have $\max_{s \in \mathbb{S}} R(\delta, s) = 1 - q_\alpha(Y_0)$.

In the Appendix, we give a proof of the statements made above. The results for the case $q_\alpha(Y_0) > 1/2$ are partly in contrast to Proposition 1(iii) where there is a *unique* minimax regret rule. However, maximal regret is the same here as in Proposition 1(iii). We have not yet generalized the results to arbitrary sample sizes $N > 0$ and to the case $F_{Y_0}(q_\alpha(Y_0)) = \alpha$.

2.3. Finite Sample Simulation

For the case of “testing an innovation,” Proposition 1(iii) establishes that the minimax regret criterion when applied to α -quantiles does not favor data-driven rules.

In this section, we conduct a simulation experiment to juxtapose the pointwise (in s) regret of the data-driven empirical success rule δ^{ES} , defined (in the case of “testing an innovation”) by

$$\delta^{ES}(w) = I(q_\alpha(Y_1, w) > q_\alpha(Y_0)) + .5I(q_\alpha(Y_1, w) = q_\alpha(Y_0)), \quad (2.25)$$

where $q_\alpha(Y_1, w)$ denotes the α -sample quantile of Y_1 for the sample w , with the regret of the minimax regret rule $\delta^1 \equiv 1$ (in the case when $q_\alpha(Y_0) \leq 1/2$), $\delta^{0.5} \equiv 1/2$ (in the case, when $q_\alpha(Y_0) = 1/2$), and the minimax regret rule $\delta^0 \equiv 0$ (in the case, when $q_\alpha(Y_0) \geq 1/2$).

Obviously, when simulating regret we cannot possibly include all states of nature $s \in \mathbb{S}$. For the simulation experiment, we create a “sufficiently rich” subset of distributions for (Y_0, Y_1) . Namely, for some $n, w \in \mathbb{N}$, we consider the set of states of nature $\mathbb{S}^E = \mathbb{S}^E(n, w)$ that consists of all discrete distributions s for Y_1 supported on a grid

$$\{0, 1/n, 2/n, \dots, n/n\} \quad (2.26)$$

with probabilities

$$P_s(Y_1 = j/n) \quad (2.27)$$

being of the form i/w , $i = 0, \dots, w$; while for the distribution of Y_0 , we consider two choices, namely,

I) $Y_0 \in \{0, q_\alpha(Y_0)\}$ with $P(Y_0 = 0) = \alpha - \varepsilon$ and $P(Y_0 = q_\alpha(Y_0)) = 1 - (\alpha - \varepsilon)$ for $\varepsilon = 0.000001$, and

II) Y_0 being continuously distributed on $[0, 1]$ with density $f(x)$ equal to $\alpha/q_\alpha(Y_0)$ for $x \leq q_\alpha(Y_0)$ and equal to $(1 - \alpha)/(1 - q_\alpha(Y_0))$ otherwise. Note that, in both I) and II), the α -quantile of Y_0 , $q_{s, \alpha}(Y_0)$, equals $q_\alpha(Y_0)$. We report results for all choices of $\alpha \in \{0.1, 0.5, 0.9\}$, $q_\alpha(Y_0) \in \{0.1, 0.5, 0.9\}$, sample size $N = 30$, and $(n, w) = (6, 12)$. The latter leads to $\mathbb{S}^E$ having cardinality 18,564.

For the given choices of n, w , and N and for each choice of α and $q_\alpha(Y_0)$, for each state of nature $s \in \mathbb{S}^E(n, w)$ and one (of the two possible) choice of distribution for Y_0 we simulate regret for the four treatment rules δ^{ES} , δ^1 , $\delta^{0.5}$, and δ^0 by generating $R = 100K$ samples of size N by drawing i.i.d. observations of the distribution of Y_1 . We use $r = 0$ when simulating α -quantiles of the outcome distribution under the various treatment rules. We analytically calculate α -quantiles for Y_1 for $Y_{B(\delta^{0.5})}$, and likewise use the true α -quantile $q_\alpha(Y_0)$ of Y_0 when calculating regret. For a given treatment rule δ and a given state of nature s , regret is calculated as $R(\delta, s) = \max\{q_\alpha(Y_0), q_{s, \alpha}(Y_1)\} - q_{s, \alpha}(Y_{B(\delta(w))})$.

We compare the treatment rules along several dimensions, namely,

- mean regret over all 18,564 states of nature $s \in \mathbb{S}^E(n, w)$,
- maximal regret over all states of nature $s \in \mathbb{S}^E(n, w)$,
- minimal regret over all states of nature $s \in \mathbb{S}^E(n, w)$, and
- the proportion of $s \in \mathbb{S}^E(n, w)$ for which regret for the empirical success rule δ^{ES} is smaller than regret for each one of its three competitors.

TABLE 1. Maximal and mean regret over all 18,564 states of nature $s \in \mathcal{S}^E(6, 12)$ for four different treatment rules.

Case I)	δ^{ES}	δ^1	$\delta^{0.5}$	δ^0	δ^{ES}	δ^1	$\delta^{0.5}$	δ^0	δ^{ES}	δ^1	$\delta^{0.5}$	δ^0
α	0.1	0.1	0.1	0.1	0.5	0.5	0.5	0.5	0.9	0.9	0.9	0.9
$q_\alpha(Y_0) = 0.1$												
max	0.9	0.1	0.9	0.9	0.1	0.1	0.9	0.9	0.1	0.1	0.9	0.9
mean	0.08	0.04	0.14	0.1	0.01	0.01	0.37	0.37	0	0	0.08	0.75
$q_\alpha(Y_0) = 0.5$												
max	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
mean	0.43	0.35	0.36	0	0.13	0.11	0.15	0.08	0.01	0	0.07	0.35
$q_\alpha(Y_0) = 0.9$												
max	0.9	0.9	0.9	0.1	0.9	0.9	0.9	0.1	0.9	0.9	0.9	0.1
mean	0.9	0.75	0.75	0	0.8	0.44	0.37	0	0.33	0.1	0.11	0.04
Case II)	δ^{ES}	δ^1	$\delta^{0.5}$	δ^0	δ^{ES}	δ^1	$\delta^{0.5}$	δ^0	δ^{ES}	δ^1	$\delta^{0.5}$	δ^0
α	0.1	0.1	0.1	0.1	0.5	0.5	0.5	0.5	0.9	0.9	0.9	0.9
$q_\alpha(Y_0) = 0.1$												
max	0.84	0.1	0.9	0.9	0.1	0.1	0.9	0.9	0	0.1	0.9	0.9
mean	0.05	0.04	0.14	0.1	0.01	0.01	0.37	0.37	0	0	0.08	0.75
$q_\alpha(Y_0) = 0.5$												
max	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.02	0.5	0.5	0.5
mean	0.06	0.35	0.36	0	0.08	0.11	0.15	0.08	0	0	0.07	0.35
$q_\alpha(Y_0) = 0.9$												
max	0.9	0.9	0.9	0.1	0.9	0.9	0.9	0.1	0.03	0.9	0.9	0.1
mean	0	0.75	0.75	0	0.04	0.44	0.37	0	0.01	0.1	0.11	0.04

Just for clarity, in a), for each treatment rule, we sum up its regret over all 18,564 states of nature $s \in \mathbb{S}^E(n, w)$ and then report that sum divided by 18,564. For a given $s \in \mathbb{S}^E(n, w)$, in our simulations, we interpret regret of δ^{ES} as smaller than regret of another rule, say δ^0 , if the simulated regret of δ^{ES} is smaller than the simulated regret of the rule δ^0 minus a threshold of ξ . If instead the simulated regret of δ^{ES} falls into the interval $[(\text{simulated regret of } \delta^0) - \xi, (\text{simulated regret of } \delta^0) + \xi]$, we record regrets of the two rules as equal for that state of nature. We take $\xi = 0.00000001$ below. Similarly, when programming the empirical success rule in (2.25), the event “ $q_\alpha(Y_1, w) = q_\alpha(Y_0)$ ” is implemented as the simulated α -quantile of Y_1 falling into the interval $[q_\alpha(Y_0) - \xi, q_\alpha(Y_0) + \xi]$.

All reported results are rounded to the second digit after the comma and so a reported zero could be as large as 0.004.

In Tables 1 and 2, we do not include results for c) because those results turn out to be equal to zero for all treatment rules and all designs except for $\delta^{0.5}$ when $q_\alpha(Y_0) \in \{0.1, 0.9\}$ in which case minimal regret equals 0.067 for all choices of α and both choices of distributions for Y_0 .

TABLE 2. Proportion (in %) of states $s \in S^E(6, 12)$ for which regret for the empirical success rule δ^{ES} is smaller than regret of non-data rules $\delta \in \{\delta^1, \delta^{0.5}, \delta^0\}$.

Case I)	δ^1	$\delta^{0.5}$	δ^0	δ^1	$\delta^{0.5}$	δ^0	δ^1	$\delta^{0.5}$	δ^0
α	0.1	0.1	0.1	0.5	0.5	0.5	0.9	0.9	0.9
$q_\alpha(Y_0) = 0.1$									
$Prop(R(\delta^{ES}, s) < R(\delta, s))$	0	33.3	33.3	0	92.5	92.5	0	33.3	100
$Prop(R(\delta^{ES}, s) \leq R(\delta, s))$	76.5	100	56.9	97.5	97.5	95.0	100	100	100
$q_\alpha(Y_0) = 0.5$									
$Prop(R(\delta^{ES}, s) < R(\delta, s))$	0	0.5	0.5	3.0	30.8	30.8	0	31.6	91.7
$Prop(R(\delta^{ES}, s) \leq R(\delta, s))$	61.0	62.7	8.3	90.7	76.2	58.6	99.5	99.5	97.8
$q_\alpha(Y_0) = 0.9$									
$Prop(R(\delta^{ES}, s) < R(\delta, s))$	0	0	0	0	2.5	2.5	0	16.2	43.1
$Prop(R(\delta^{ES}, s) \leq R(\delta, s))$	43.3	43.4	0	21.0	11.7	2.5	66.7	66.7	43.1
Case II)	δ^1	$\delta^{0.5}$	δ^0	δ^1	$\delta^{0.5}$	δ^0	δ^1	$\delta^{0.5}$	δ^0
α	0.1	0.1	0.1	0.5	0.5	0.5	0.9	0.9	0.9
$q_\alpha(Y_0) = 0.1$									
$Prop(R(\delta^{ES}, s) < R(\delta, s))$	16.2	73.0	73.0	0.3	95.3	95.3	0	33.4	100
$Prop(R(\delta^{ES}, s) \leq R(\delta, s))$	76.5	100	73.0	100	97.5	95.3	100	100	100
$q_\alpha(Y_0) = 0.5$									
$Prop(R(\delta^{ES}, s) < R(\delta, s))$	71.2	73.4	71.6	13.6	40.7	39.8	2.2	33.8	91.7
$Prop(R(\delta^{ES}, s) \leq R(\delta, s))$	98.3	100	79.6	100	86.6	65.3	100	100	97.8
$q_\alpha(Y_0) = 0.9$									
$Prop(R(\delta^{ES}, s) < R(\delta, s))$	99.5	99.6	99.6	84.0	82.1	75.6	56.9	73.0	43.1
$Prop(R(\delta^{ES}, s) \leq R(\delta, s))$	100	100	99.6	100	95.7	75.6	100	100	43.1

Table 1 provides results for maximal and mean regret over all 18,564 states of nature $s \in S^E(6, 12)$ for the four different treatment rules. Results for rules that are minimax regret in a given setting are reported in bold.

We first discuss results for maximal regret. The treatment rules that are known to be minimax regret for unrestricted $\mathbb{S}$ also have smallest maximal regret over $\mathbb{S}^E(n, w)$ (relative to the other treatment rules considered here) for all cases except for case II) with $\alpha = 0.9$, when $q_\alpha(Y_0) = 0.1$ (in which case, δ^{ES} has smaller maximal regret than the optimal rule δ^1), when $q_\alpha(Y_0) = 0.5$ (in which case, it is known that all four rules are optimal but in finite samples for $\mathbb{S}^E(n, w)$ again δ^{ES} does best), and finally when $q_\alpha(Y_0) = 0.9$ (in which case, δ^{ES} has smaller maximal regret than the optimal rule δ^0). The explanation is of course that $\mathbb{S}^E(n, w)$ does not contain those states of nature that would inflict the highest regret on δ^{ES} (like the particular Bernoulli and discrete distributions for Y_0 and Y_1 , respectively, that are used in the proof of Proposition 1(iii)). Furthermore, reported finite sample maximal regret over $\mathbb{S}^E(n, w)$ for the optimal rules matches

the theoretical value $\min\{q_\alpha(Y_0), 1 - q_\alpha(Y_0)\}$ reported in Proposition 1(iii) except for case II) with $\alpha = 0.9$ when $q_\alpha(Y_0) = 0.5$ where the optimal rule δ^{ES} has maximal regret smaller than 0.5 (which occurs, again, because $\mathbb{S}^E(n, w)$ is not rich enough). An open question from Proposition 1(iii) is whether for $q_\alpha(Y_0) < 0.5$ other minimax regret rules besides δ^1 might exist. The simulations for $q_\alpha(Y_0) = 0.1$ are compatible with the possibility that when $\alpha = 0.5$ or 0.9 also δ^{ES} might be minimax regret.

We next discuss results for mean regret. In most cases where $q_\alpha(Y_0) \in \{0.1, 0.9\}$ in which case either δ^0 or δ^1 are minimax regret, their mean performance is also best (or very close to best) among the four treatment rules. On the other hand, when $q_\alpha(Y_0) = 0.5$ (in which case all four treatment rules are minimax regret according to Proposition 1(iii)) we see huge differences in mean performance across the four treatment rules for a given case I) or II) and α , but also huge differences in performance for a given treatment rule and α across cases I) and II). With respect to the former point, in case I), when $\alpha = 0.1$, mean regret for the four rules is in the interval $[0, 0.43]$. With regards to the latter point, for example, for δ^{ES} , when $\alpha = 0.1$, mean regret equals 0.43 and 0.06, in cases I) and II), respectively. Differences across cases I) and II) are also often huge for other quantiles. For example, again for δ^{ES} , when $q_\alpha(Y_0) = 0.9$ and $\alpha = 0.1$, mean regret equals 0.9 and 0, in cases I) and II), respectively. (Recall that we round results to the second digit. When $q_\alpha(Y_0) = 0.9$ and $\alpha = 0.1$, there are not many distributions for Y_1 in $\mathbb{S}^E(n, w)$ that have an α -quantile that exceeds 0.9.)

We next discuss the results for exercise d) contained in Table 2 where we report the proportion of the 18,564 states of nature $s \in S^E(6, 12)$ for which the regret of δ^{ES} is smaller than the regret of δ where we consider all δ from the set of no-data rules $\{\delta^1, \delta^{0.5}, \delta^0\}$. The results indicate that for many states s , choices of $q_\alpha(Y_0)$, α , and the distribution for Y_0 , $R(\delta^{ES}, s)$, and $R(\delta, s)$ are very close which leads to numerically unstable results. To deal with the instability, we introduce the buffer ξ as explained above and $\text{Prop}(R(\delta^{ES}, s) < R(\delta, s))$ and $\text{Prop}(R(\delta^{ES}, s) \leq R(\delta, s))$ in Table 2 represent the proportion of states s for which $R(\delta^{ES}, s) < R(\delta, s) - \xi$ and $R(\delta^{ES}, s) < R(\delta, s) + \xi$, respectively. Those two proportions can be drastically different, suggesting that, in many scenarios, regrets for δ^{ES} and δ are (virtually) identical. According to the measure $\text{Prop}(R(\delta^{ES}, s) \leq R(\delta, s))$, maybe somewhat surprisingly given the results from Table 1, δ^{ES} is to be preferred over the other three rules in all scenarios considered in Case II) (except the case $\alpha = q_\alpha(Y_0) = 0.9$ for δ^0) and in the majority of scenarios in Case I) (except compared to δ^0 when $q_\alpha(Y_0) = 0.5$, $\alpha = 0.1$ and except for most cases with $q_\alpha(Y_0) = 0.9$ and all three rules). Quite often $\text{Prop}(R(\delta^{ES}, s) \leq R(\delta, s))$ is reported as higher than 95%, but the improvement in regret for many states of nature is minuscule. For example, compared to $\delta^{0.5}$ in Case I) with $\alpha = q_\alpha(Y_0) = 0.1$ only in 33.3% of the cases $(R(\delta^{ES}, s) < R(\delta, s))$ while for 100% of the cases $(R(\delta^{ES}, s) \leq R(\delta, s))$ implying that the regret of δ^{ES} in 66.7% of the cases is at most ξ smaller than the regret of $\delta^{0.5}$.

3. TREATMENT CHOICE WITH A COVARIATE

As in Stoye (2009) we next allow for a discrete covariate $X \in \mathcal{X} = \{x_1, \dots, x_K\}$ that is observed both in the sample and in the treatment data. Outcomes $Y_{t,x}$ now carry a double subindex to indicate the treatment status $t \in \{0, 1\}$ and the value of the covariate $x \in \mathcal{X}$. It is assumed that x_k occurs with positive probability for each $k = 1, \dots, K$. Denote by F_X the distribution of X . A state of the world $s \in \mathbb{S}$ now represents a joint distribution for $(Y_{t,x})_{t \in \{0, 1\}, x \in \mathcal{X}}$.

A sample $w = w_N$ now consists of realizations (t_i, x_i, y_i) for $i = 1, \dots, N$ of $(T, X, Y_{T,X})$, where y_i is a realization of Y_{t_i, x_i} , and we consider again sampling designs (i)–(iii) from Section 2.2. It is assumed that $x_i, i = 1, \dots, N$, are i.i.d. and F_X is independent of s and T .

In design (i), “fixed number of treated/untreated units,” there are N_0 observations $(0, x_i, y_i)$ with y_i being i.i.d. draws of $Y_{0, x_i}, i = 1, \dots, N_0$ and $N_1 = N - N_0$ observations $(1, x_i, y_i)$ with y_i being i.i.d. draws of $Y_{1, x_i}, i = N_0 + 1, \dots, N$. Note that there are not typically equally many treated and untreated observations for each covariate $x_k, k = 1, \dots, K$. In fact, there may be zero observations altogether in the sample for a given x_k .

In design (ii), “random sampling,” the observations (t_i, x_i, y_i) for $i = 1, \dots, N$ are i.i.d. realizations of $(T, X, Y_{T,X})$ with $P(T_i = 1) = p \in (0, 1)$ with T, X , and the potential outcomes (whose distribution is s) being independent of each other.

In design (iii), “testing an innovation,” $(1, x_i, y_i)$ for $i = 1, \dots, N$ are observed, where y_i is an independent realization of Y_{1, x_i} , for $i = 1, \dots, N$.

In design (iii), we assume the DM knows the α -quantile, denoted by $q_\alpha(Y_{0,X})$, of $Y_{0,X}$ and nature can choose from joint distributions $s \in \mathbb{S}$ for $(Y_{t,x})_{t \in \{0, 1\}, x \in \mathcal{X}}$ such that the α -quantile of $Y_{0,X}$ equals $q_\alpha(Y_{0,X})$.

A treatment rule δ maps a sample w_N onto treatment probabilities for each x_k for $k = 1, \dots, K$, that is, $\delta(w_N) \in [0, 1]^K$, where the k -th component of $\delta(w_N)$ indicates the treatment probability for individuals with covariate x_k for $k = 1, \dots, K$. Denote the k -th component of $\delta(w_N)$ by $\delta_{x_k}(w_N)$ for $k = 1, \dots, K$. With some abuse of notation, for each design (i)–(iii), we denote the set of all treatment rules by the same symbol $\mathbb{D}$ (even though it means something different for different designs).

The object of interest is

$$u(\delta, s) = q_{s, F_X, \alpha}(Y_{B(\delta_X(w_N)), X}) \quad (3.1)$$

the α -quantile of the outcome distribution. With that definition, regret is then defined formally analogously to the setup without a covariate, namely, $R(\delta, s) = \sup_{d \in \mathbb{D}} u(d, s) - u(\delta, s)$. Alternatively, one could focus on the α -quantile of the outcome distribution for a particular covariate only, x_1 say, $u_{x_1}(\delta, s) = q_{s, F_X, \alpha}(Y_{B(\delta_{x_1}(w_N)), x_1})$ and obtain analogous results to the ones in Corollary 2 below.

If the space of probability distributions for nature $\mathbb{S}$ is unrestricted and thus equals the space of all distributions for $(Y_{t,x})_{t \in \{0, 1\}, x \in \mathcal{X}}$ one can show in designs (i) and (ii) (as an implication of the proof of Proposition 1) that for every treatment rule δ maximal regret over $\mathbb{S}$ equals 1, $\max_{s \in \mathbb{S}} R(\delta, s) = 1$. This result then

immediately implies that every treatment rule is minimax regret. The corollary (to Proposition 1) that follows gives a stronger result; it shows that maximal regret continues to be 1 even if certain restrictions are imposed on $\mathbb{S}$. For simplicity of the presentation, assume $\alpha \in (0, 1)$.

COROLLARY 2. (i) and (ii). *In the case of the designs “fixed number of treated/untreated units” and “random sampling” for any treatment rule $\delta \in \mathbb{D}$, $\max_{s \in \mathbb{S}} R(\delta, s) = 1$ if $\mathbb{S}$ includes as a subset all joint distributions for $(Y_{t,x})_{t \in \{0,1\}, x \in \mathcal{X}}$ whose marginals are independent Bernoulli distributions. Therefore, any $\delta \in \mathbb{D}$ is minimax regret.*

(iii) *In the case of “testing an innovation” assume the DM knows the α -quantile, denoted by $q_\alpha(Y_{0,x})$, of $Y_{0,x}$. If $q_\alpha(Y_{0,x}) = 1/2$, then any $\delta \in \mathbb{D}$ is minimax regret; if $q_\alpha(Y_{0,x}) > 1/2$, then $\delta^0 = 0$ is the unique minimax regret rule, and if $q_\alpha(Y_{0,x}) < 1/2$, then $\delta^1 = 1$ is a minimax regret rule. In each case, $\max_{s \in \mathbb{S}} R(\delta, s) = \min\{q_\alpha(Y_{0,x}), 1 - q_\alpha(Y_{0,x})\}$.*

Comments: 1. Various variants could be considered in design (iii). For example, one could instead assume that the joint distribution $(Y_{0,x})_{x \in \mathcal{X}}$ is known (in which case nature only chooses a joint distribution for $(Y_{1,x})_{x \in \mathcal{X}}$) or one could assume that the DM knows the vector $(q_\alpha(Y_{0,x_1}), \dots, q_\alpha(Y_{0,x_K}))$ of α -quantiles of all the marginal distributions $(Y_{0,x})_{x \in \mathcal{X}}$ and nature can choose from joint distributions $s \in \mathbb{S}$ for $(Y_{t,x})_{t \in \{0,1\}, x \in \mathcal{X}}$ such that all the marginals $(Y_{0,x})_{x \in \mathcal{X}}$ have the required α -quantiles.

2. One could also consider alternative sampling designs (i)–(iii), where in the sampling stage, rather than being randomly assigned, the values of the covariate X are assigned deterministically, as is done for treatment status in design (i). This could be referred to as “stratified sampling.”

For brevity, we do not explicitly deal with these variations.

4. CONCLUSION

We derive minimax regret treatment rules in finite samples when an α -quantile of the outcome distribution is the focus of interest. We establish that when the sample i) consists of a fixed number of untreated/treated units or ii) is generated via random treatment assignment then *all* treatment rules are minimax regret and therefore the minimax regret criterion is not helpful in singling out a recommended treatment rule. Given that the same shortcoming applies to the max–min criterion, an important question concerns finding a meaningful criterion in this setup based on which an optimal treatment rule should be chosen. The idea from Montiel Olea et al. (2026) to look for rules that randomize “the least” in a situation where there are multiple minimax regret rules would not lead to a unique rule in our setup because in cases i) and ii) both δ^0 and δ^1 are minimax regret and never randomize. Given all these facts, it seems reasonable to simply adopt a rule that is minimax regret optimal when regret is based on the notion of *expected* welfare (in particular, such a rule is optimal according to criterion (2.7)).

We also establish that when iii) the sample consists of only realizations from the treated population while the α -quantile of the untreated population is known, never treating is the unique minimax rule if the known quantile exceeds 0.5, while always treating is a minimax rule if that quantile is strictly smaller than 0.5, and finally, if the quantile equals 0.5, then any treatment rule is minimax regret. It follows that based on the minimax regret criterion, rules that take the data into consideration are never strictly preferred.

An interesting but quite difficult extension concerns applying the minimax regret criterion in finite samples to CVaR, defined as $S_{s,\alpha}(Y) := \alpha^{-1}E_s[Y1(Y \leq q_{s,\alpha}(Y))]$ for a random variable Y rather than to the α -quantile of Y .⁹ Another extension involves modifying the analysis in Huang and Guggenberger (2025) that considers pilot data from a setup where the objective is expected outcome to one where the objective is a quantile.

A. APPENDIX

A.1. Proof of Proposition 1

We use the notation $P_s(A)$ to denote the probability of the event A when nature chooses the state of the world $s \in \mathbb{S}$ and we write shorthand w instead of w_N for the sample of size N . Note that, for a state of the world $s \in \mathbb{S}$ and treatment rule $\delta \in \mathbb{D}$,

$$\begin{aligned} P_s(Y_{B(\delta(w))} \leq q) \\ &= P_s(B(\delta(w)) = 1 \text{ \& } Y_1 \leq q) + P_s(B(\delta(w)) = 0 \text{ \& } Y_0 \leq q) \\ &= P_s(B(\delta(w)) = 1)P_s(Y_1 \leq q) + P_s(B(\delta(w)) = 0)P_s(Y_0 \leq q) \end{aligned} \quad (\text{A.1})$$

and analogously for $P_s(Y_{B(\delta(w))} \geq q)$, where the second equality uses independence. By definition, an α -quantile q of $Y_{B(\delta(w))}$ satisfies $P_s(Y_{B(\delta(w))} \leq q) \geq \alpha$ and $P_s(Y_{B(\delta(w))} \geq q) \geq 1 - \alpha$.

The proofs of (i) and (ii) proceed by showing that for any treatment rule $\delta \in \mathbb{D}$, there exists a state of the world $s_\delta \in \mathbb{S}$ for which $R(\delta, s_\delta) = 1$. Because it is also true that for any treatment rule $\delta \in \mathbb{D}$, $\max_{s \in \mathbb{S}} R(\delta, s) \leq 1$, it follows that any treatment rule is minimax.

Lemma 3(ii) below establishes the (unsurprising) result that for any arbitrary $s \in \mathbb{S}$,

$$\max_{d \in \mathbb{D}} u(d, s) = \max_{d \in \mathbb{D}} q_{s,\alpha}(Y_{B(d(w))}) = \max\{q_{s,\alpha}(Y_0), q_{s,\alpha}(Y_1)\}. \quad (\text{A.2})$$

This result implies a simplified formula for $R(\delta, s)$ that we will use from now on.

In the proof of (i) and (ii) that follows, for any given treatment rule $\delta \in \mathbb{D}$, we will construct an $s_\delta \in \mathbb{S}$ for which $\max\{q_{s_\delta,\alpha}(Y_0), q_{s_\delta,\alpha}(Y_1)\} = 1$ and $u(\delta, s_\delta) = 0$. Namely, throughout the proof of (i) and (ii) for an arbitrary treatment rule $\delta \in \mathbb{D}$, we construct an $s_\delta \in \mathbb{S}$ whose independent marginals for Y_0 and Y_1 are Bernoulli (supported on $\{0, 1\}$) and for which $R(\delta, s_\delta) = 1$. Under these restrictions, the distribution s_δ for (Y_0, Y_1) is then fully

⁹Or, using the more general definition $S_{s,\alpha}(Y) := \sup_{\gamma \in \mathbb{R}} \{\gamma - \alpha^{-1}E_s[\gamma - Y]_+\}$, see Qi et al. (2023), where $[Y]_+ = \max\{Y, 0\}$ denotes the positive part of Y .

defined by

$$a_0 := P_{s_\delta}(Y_0 = 0) \text{ and } a_1 := P_{s_\delta}(Y_1 = 0). \quad (\text{A.3})$$

The α -quantiles of Y_0 , Y_1 , and $Y_{B(\delta(w))}$ in the constructions below will be unique when $\alpha \in (0, 1)$. Therefore, the particular value of $r \in [0, 1]$ has no impact on the results.

Part (i). As in (A.1) and with s_δ just defined

$$\begin{aligned} P_{s_\delta}(Y_{B(\delta(w))} = 0) &= \sum_{u \in \{0, 1\}^{N_0}} \sum_{v \in \{0, 1\}^{N_1}} a_0^{N_0 - \bar{u}} (1 - a_0)^{\bar{u}} a_1^{N_1 - \bar{v}} (1 - a_1)^{\bar{v}} \\ &\quad [\delta((u', v')') a_1 + (1 - \delta((u', v')')) a_0] \\ &= \sum_{i=0}^{N_0} \sum_{j=0}^{N_1} a_0^{N_0-i} (1 - a_0)^i a_1^{N_1-j} (1 - a_1)^j \left[\delta^{i,j} a_1 + \left(\binom{N_0}{i} \binom{N_1}{j} - \delta^{i,j} \right) a_0 \right], \end{aligned} \quad (\text{A.4})$$

where for a vector $u \in \{0, 1\}^{N_0}$, we denote by $\bar{u} \in \mathbb{N}$ the sum of its components (i.e., the number of 1's in the vector) and similarly for other vectors, and

$$\delta^{i,j} := \sum_{\substack{u \in \{0, 1\}^{N_0}, v \in \{0, 1\}^{N_1} \\ \bar{u}=i \text{ and } \bar{v}=j}} \delta((u', v')'). \quad (\text{A.5})$$

Note that in the second line of (A.4), the term $a_0^{N_0 - \bar{u}} (1 - a_0)^{\bar{u}} a_1^{N_1 - \bar{v}} (1 - a_1)^{\bar{v}}$ is the probability to observe particular vectors u and v while the term $\delta((u', v')') a_1 + (1 - \delta((u', v')')) a_0$ is the probability that an outcome 0 is reached when the sample consists of u and v .

Assume $\alpha \in (0, 1)$ first. When in the definition of s_δ in (A.3), we take $(a_0, a_1) = (1, \alpha)$ we obtain from (A.4),

$$\begin{aligned} P_{s_\delta}(Y_{B(\delta(w))} = 0) &= \sum_{j=0}^{N_1} \alpha^{N_1-j} (1 - \alpha)^j \left[\delta^{0,j} \alpha + \left(\binom{N_1}{j} - \delta^{0,j} \right) \right] \\ &= \sum_{j=0}^{N_1} [\alpha^{N_1-j} (1 - \alpha)^j (\alpha - 1)] \delta^{0,j} + \sum_{j=0}^{N_1} \alpha^{N_1-j} (1 - \alpha)^j \binom{N_1}{j}. \end{aligned} \quad (\text{A.6})$$

Given that the coefficient $\alpha^{N_1-j} (1 - \alpha)^j (\alpha - 1)$ in front of $\delta^{0,j}$ is negative (noting that $\alpha \in (0, 1)$), it follows that $P_{s_\delta}(Y_{B(\delta(w))} = 0)$ is strictly decreasing in each $\delta^{0,j}$. Therefore, we obtain

$$P_{s_\delta}(Y_{B(\delta(w))} = 0) \geq \sum_{j=0}^{N_1} \alpha^{N_1-j} (1 - \alpha)^j \alpha \binom{N_1}{j} = \alpha, \quad (\text{A.7})$$

where the right-hand bound follows from (A.6) by replacing $\delta^{0,j}$ by its maximal value $\binom{N_1}{j}$.

Consider first the case where at least one of the $\delta^{0,j}$ (for $j = 0, \dots, N_1$) is smaller than $\binom{N_1}{j}$. In that case, it follows that $P_{s_\delta}(Y_{B(\delta(w))} = 0) > \alpha$. Furthermore, notice that $P_{s_\delta}(Y_{B(\delta(w))} = 0)$ is continuous in a_1 and therefore $P_{s_\delta}(Y_{B(\delta(w))} = 0) > \alpha$ when instead of $(a_0, a_1) = (1, \alpha)$, s_δ is taken as $(a_0, a_1) = (1, \alpha - \varepsilon)$ for some small enough $\varepsilon > 0$. But then for that choice of s_δ , $u(\delta, s_\delta) = 0$ and $\max\{q_{s_\delta, \alpha}(Y_0), q_{s_\delta, \alpha}(Y_1)\} = q_{s_\delta, \alpha}(Y_1) = 1$. Therefore, $R(\delta, s_\delta) = 1$ as desired.

Next, consider the case where $\delta^{0,j} = \binom{N_1}{j}$ for all $j = 0, \dots, N_1$ (which implies that $\delta((0^{N_0'}, v')') = 1$ for all $v \in \{0, 1\}^{N_1}$, where 0^{N_0} denotes an N_0 -dimensional column vector

of zeros). In that case, from (A.4), we obtain with $(a_0, a_1) = (\alpha, 1)$

$$\begin{aligned}
 P_{s_\delta}(Y_{B(\delta(w))} = 0) &= \sum_{i=0}^{N_0} \alpha^{N_0-i} (1-\alpha)^i \left[\delta^{i,0} + \left(\binom{N_0}{i} - \delta^{i,0} \right) \alpha \right] \\
 &= \alpha^{N_0} + \sum_{i=1}^{N_0} \alpha^{N_0-i} (1-\alpha)^i \left[\delta^{i,0} (1-\alpha) + \binom{N_0}{i} \alpha \right] \\
 &\geq \alpha^{N_0} + \alpha \sum_{i=1}^{N_0} \alpha^{N_0-i} (1-\alpha)^i \binom{N_0}{i} \\
 &= \alpha^{N_0} + \alpha(1 - \alpha^{N_0}) \\
 &> \alpha,
 \end{aligned} \tag{A.8}$$

where the second equality uses $\delta^{0,0} = \binom{N_1}{0} = 1$, the first inequality follows from setting $\delta^{i,0} = 0$ for $i = 1, \dots, N_0$, and the second inequality follows from $\alpha \in (0, 1)$. The remainder of the proof is as above, namely, by continuity $P_{s_\delta}(Y_{B(\delta(w))} = 0) > \alpha$ for an s_δ with $(a_0, a_1) = (\alpha - \varepsilon, 1)$ for a small enough $\varepsilon > 0$. For such s_δ , regret equals 1.

Now consider the case $\alpha = 0$. Take any δ and define s_δ by setting $a_0 = P_{s_\delta}(Y_0 = 0) = 0$ and $a_1 = P_{s_\delta}(Y_1 = 0) = \varepsilon > 0$. Then, $q_{s_\delta,0}(Y_1) = 0$ and any $q \in [0, 1]$ satisfies the condition in (2.4) for $X = Y_0$. Thus, with $r = 1$, the 0-quantile of Y_0 equals 1. Thus, in order for regret to equal 1 under s_δ , it is enough to establish that the 0-quantile of $Y_{B(\delta(w))}$ equals 0. The latter follows if $P_{s_\delta}(Y_{B(\delta(w))} = 0) > 0$. From (A.4),

$$P_{s_\delta}(Y_{B(\delta(w))} = 0) = \varepsilon \sum_{j=0}^{N_1} \varepsilon^{N_1-j} (1-\varepsilon)^j \delta^{N_0,j} \tag{A.9}$$

with $\delta^{N_0,j} = \sum_{v \in \{0,1\}^{N_1}, \bar{v}=j} \delta((1, \dots, 1, v')')$. If $P_{s_\delta}(Y_{B(\delta(w))} = 0) > 0$, the proof is complete. If not, and $P_{s_\delta}(Y_{B(\delta(w))} = 0) = 0$, it must be that $\delta((1, \dots, 1, v')') = 0$ for all $v \in \{0, 1\}^{N_1}$. If $\delta((1, \dots, 1, v')') = 0$ for all $v \in \{0, 1\}^{N_1}$, define s'_δ by setting $a_0 = P_{s'_\delta}(Y_0 = 0) = \varepsilon$ and $a_1 = P_{s'_\delta}(Y_1 = 0) = 0$. Then, by (A.4), with $\delta^{i,N_1} = \sum_{u \in \{0,1\}^{N_0}, \bar{u}=i} \delta((u', 1, \dots, 1)')$ $\leq \binom{N_0}{i}$,

$$P_{s'_\delta}(Y_{B(\delta(w))} = 0) = \varepsilon \sum_{i=0}^{N_0} \varepsilon^{N_0-i} (1-\varepsilon)^i \left(\binom{N_0}{i} - \delta^{i,N_1} \right). \tag{A.10}$$

But $1 - \delta((1, \dots, 1, v')') = 1$ for all $v \in \{0, 1\}^{N_1}$ implies $\binom{N_0}{i} - \delta^{i,N_1} = 1$ when $i = N_0$. Thus, by (A.10), $P_{s'_\delta}(Y_{B(\delta(w))} = 0) > 0$. But, the latter implies that regret equals 1 under s'_δ .

Finally, consider the case $(\alpha = 1 \text{ and } r = 0)$. Consider a treatment rule δ that is not equal to 1 wp1 and not equal to 0 wp1. We will show that $\max_{s \in \mathbb{S}} R(\delta, s) = 0$. To see this, fix $s \in \mathbb{S}$. By the definition of an α -quantile in (2.4) and the choice $r = 0$ note that, for any $q < q_{s,1}(Y_0)$, it must be that $P_s(Y_0 \leq q) < 1$, and likewise for any $q < q_{s,1}(Y_1)$, it must be that $P_s(Y_1 \leq q) < 1$. Therefore, using (A.1), for any $q < \max\{q_{s,1}(Y_0), q_{s,1}(Y_1)\}$, we have

$$P_s(Y_{B(\delta(w))} \leq q) = P_s(B(\delta(w)) = 1)P_s(Y_1 \leq q) + P_s(B(\delta(w)) = 0)P_s(Y_0 \leq q) < 1,$$

which implies $q < q_{s,1}(Y_{B(\delta(w))})$. By Lemma 3, $q_{s,1}(Y_{B(\delta(w))}) \leq \max\{q_{s,1}(Y_0), q_{s,1}(Y_1)\}$ and thus $q_{s,1}(Y_{B(\delta(w))}) = \max\{q_{s,1}(Y_0), q_{s,1}(Y_1)\}$. But that implies that $R(\delta, s) = 0$ and because $s \in \mathbb{S}$ that proves the desired result.

Part (ii). The proof of (ii) is similar to case (i). Again we start off with $\alpha \in (0, 1)$ and define s_δ as in (A.3). We obtain

$$\begin{aligned}
 P_{s_\delta}(Y_{B(\delta(w))} = 0) &= \sum_{\substack{t=(t_1, \dots, t_N)' \in \{0, 1\}^N, \\ y=(y_{t_1}, \dots, y_{t_N})' \in \{0, 1\}^N}} p^{\bar{t}}(1-p)^{N-\bar{t}} a_0^{N_{0ty}} (1-a_0)^{N-\bar{t}-N_{0ty}} a_1^{N_{1ty}} (1-a_1)^{\bar{t}-N_{1ty}} \\
 &\quad \times [\delta((t, y))a_1 + (1 - \delta((t, y)))a_0] \\
 &= \sum_{\substack{T=0, \dots, N, \\ i=0, \dots, N-T, \\ j=0, \dots, T}} p^T(1-p)^{N-T} a_0^i (1-a_0)^{N-T-i} a_1^j (1-a_1)^{T-j} \\
 &\quad \times [\delta^{T, i, j} a_1 + \left(\binom{N}{T} \binom{N-T}{i} \binom{T}{j} - \delta^{T, i, j} \right) a_0] \\
 &= \sum_{\substack{T=0, \dots, N, \\ i=0, \dots, N-T, \\ j=0, \dots, T}} p^T(1-p)^{N-T} a_0^i (1-a_0)^{N-T-i} a_1^j (1-a_1)^{T-j} \\
 &\quad \times [(a_1 - a_0) \delta^{T, i, j} + \left(\binom{N}{T} \binom{N-T}{i} \binom{T}{j} \right) a_0], \tag{A.11}
 \end{aligned}$$

where for given vectors $t \in \{0, 1\}^N$ and $y \in \{0, 1\}^N$, N_{0ty} denotes the number of y_{t_i} , $i = 1, \dots, N$, for which $y_{t_i} = 0$ and $t_i = 0$, N_{1ty} is the number of y_{t_i} , $i = 1, \dots, N$, for which $y_{t_i} = 0$ and $t_i = 1$, where again $\bar{t} \in \mathbb{N}$ for a vector $t \in \{0, 1\}^N$ denotes the sum of its components, and

$$\delta^{T, i, j} := \sum_{\substack{t=(t_1, \dots, t_N)' \in \{0, 1\}^N, \\ y=(y_{t_1}, \dots, y_{t_N})' \in \{0, 1\}^N, \\ \bar{t}=T, N_{0ty}=i, N_{1ty}=j}} \delta((t, y)). \tag{A.12}$$

Note that $N_{0ty} \leq N - \bar{t}$ and $N_{1ty} \leq \bar{t}$. In the second line of (A.11), the factor $p^{\bar{t}}(1-p)^{N-\bar{t}}$ captures the probability of observing exactly $\bar{t}$ treatments and $N - \bar{t}$ controls in the sample, while the factor $a_0^{N_{0ty}}(1-a_0)^{N-\bar{t}-N_{0ty}}$ captures the probability that among the $N - \bar{t}$ observations from the control group exactly N_{0ty} zeros are observed, while, finally, the factor $a_1^{N_{1ty}}(1-a_1)^{\bar{t}-N_{1ty}}$ captures the probability that among the $\bar{t}$ observations from treated individuals exactly N_{1ty} zeros are observed.

Evaluating $P_{s_\delta}(Y_{B(\delta(w))} = 0)$ when s_δ takes $(a_0, a_1) = (1, \alpha)$, we obtain

$$\begin{aligned}
 P_{s_\delta}(Y_{B(\delta(w))} = 0) &= \sum_{\substack{T=0, \dots, N, \\ j=0, \dots, T}} p^T(1-p)^{N-T} \alpha^j (1-\alpha)^{T-j} \\
 &\quad [(\alpha - 1) \delta^{T, N-T, j} + \left(\binom{N}{T} \binom{T}{j} \right)]. \tag{A.13}
 \end{aligned}$$

Because $(\alpha - 1) < 0$ and $\alpha > 0$, $P_{s_\delta}(Y_{B(\delta(w))} = 0)$ is strictly decreasing in $\delta^{T, N-T, j}$ for all $T = 0, \dots, N$ and $j = 0, \dots, T$ and is minimized when $\delta^{T, N-T, j}$ takes on its maximal value

$\binom{N}{T} \binom{T}{j}$. Therefore,

$$\begin{aligned} P_{s_\delta}(Y_{B(\delta(w))} = 0) &\geq \sum_{T=0, \dots, N} p^T (1-p)^{N-T} \alpha^j (1-\alpha)^{T-j} \binom{N}{T} \binom{T}{j} \\ &= \alpha \sum_{T=0, \dots, N} \binom{N}{T} p^T (1-p)^{N-T} \sum_{j=0, \dots, T} \binom{T}{j} \alpha^j (1-\alpha)^{T-j} \\ &= \alpha. \end{aligned} \quad (\text{A.14})$$

If $\delta^{T, N-T, j} < \binom{N}{T} \binom{T}{j}$ for any $T = 0, \dots, N$ and $j = 0, \dots, T$, then by an argument as in part (i), it follows that an s_δ with $(a_0, a_1) = (1, \alpha - \varepsilon)$ for $\varepsilon > 0$ small enough leads to regret of 1. This holds because for that s_δ we have $q_{s_\delta, \alpha}(Y_0) = 0$, $q_{s_\delta, \alpha}(Y_1) = 1$, and $q_{s_\delta, \alpha}(Y_{B(\delta(w))}) = 0$.

If on the other hand, $\delta^{T, N-T, j} = \binom{N}{T} \binom{T}{j}$ for all $T = 0, \dots, N$ and $j = 0, \dots, T$, then, as in part (i), one can show that s_δ defined by $(a_0, a_1) = (\alpha - \varepsilon, 1)$ for a small enough $\varepsilon > 0$ leads to regret of 1. Namely, for s_δ with $(a_0, a_1) = (\alpha, 1)$, we get from (A.11)

$$\begin{aligned} P_{s_\delta}(Y_{B(\delta(w))} = 0) &= \sum_{T=0, \dots, N} p^T (1-p)^{N-T} \sum_{i=0, \dots, N-T} \alpha^i (1-\alpha)^{N-T-i} \\ &\quad [(1-\alpha)\delta^{T, i, T} + \binom{N}{T} \binom{N-T}{i} \alpha] \\ &= \sum_{T=0, \dots, N} p^T (1-p)^{N-T} \\ &\quad \times \{\alpha^{N-T} \binom{N}{T} + \sum_{i=0, \dots, N-T-1} \alpha^i (1-\alpha)^{N-T-i} [(1-\alpha)\delta^{T, i, T} + \binom{N}{T} \binom{N-T}{i} \alpha]\} \\ &\geq \sum_{T=0, \dots, N} \binom{N}{T} p^T (1-p)^{N-T} \{\alpha^{N-T} + \alpha \sum_{i=0, \dots, N-T-1} \binom{N-T}{i} \alpha^i (1-\alpha)^{N-T-i}\} \\ &= \sum_{T=0, \dots, N} \binom{N}{T} p^T (1-p)^{N-T} \{\alpha^{N-T} + \alpha(1-\alpha^{N-T})\} \\ &> \alpha \sum_{T=0, \dots, N} \binom{N}{T} p^T (1-p)^{N-T} \\ &= \alpha, \end{aligned} \quad (\text{A.15})$$

where the second equality simply takes the last summand with index $i = N - T$ out of the second summation sign and uses $\delta^{T, N-T, T} = \binom{N}{T}$, the inequality is obtained by setting $\delta^{T, i, T} = 0$ for all $T = 0, \dots, N$ and $i = 0, \dots, N - T - 1$, and the strict inequality uses $\alpha^{N-T} + \alpha(1-\alpha^{N-T}) > \alpha$. By an argument as in part (i), (A.15) implies that for s'_δ with $(a_0, a_1) = (\alpha - \varepsilon, 1)$ with sufficiently small $\varepsilon > 0$ regret of 1 can be obtained because, under s'_δ , we have $q_{s'_\delta, \alpha}(Y_0) = 1$, $q_{s'_\delta, \alpha}(Y_1) = 0$, and $q_{s'_\delta, \alpha}(Y_{B(\delta(w))}) = 0$.

Next, assume $\alpha = 0$. Define s_δ such that $a_0 = P_{s_\delta}(Y_0 = 0) = 0$ and $a_1 = P_{s_\delta}(Y_1 = 0) = \varepsilon > 0$. That and $r = 1$ imply that $q_{s_\delta, \alpha}(Y_0) = 1$ and $q_{s_\delta, \alpha}(Y_1) = 0$. Then, from (A.11),

$$\begin{aligned} P_{s_\delta}(Y_{B(\delta(w))} = 0) &= \sum_{\substack{t=(t_1, \dots, t_N)' \in \{0, 1\}^N, \\ y=(y_1, \dots, y_N)' \in \{0, 1\}^N, \\ y_j=1 \text{ if } t_j=0 \text{ for } j=1, \dots, N}} p^{\bar{t}} (1-p)^{N-\bar{t}} \varepsilon^{N_{1ty}} (1-\varepsilon)^{\bar{t}-N_{1ty}} \delta((t, y)) \varepsilon. \end{aligned} \quad (\text{A.16})$$

Thus, $P_{s_\delta}(Y_{B(\delta(w))} = 0) > 0$ (which implies $R(\delta, s_\delta) = 1$), unless $\delta((t, y)) = 0$ for all vectors $t, y \in \{0, 1\}^N$ such that $y_{t_j} = 1$ if $t_j = 0$ for $j = 1, \dots, N$. For such δ consider instead s'_δ such that $a_0 = P_{s'_\delta}(Y_0 = 0) = \varepsilon > 0$ and $a_1 = P_{s'_\delta}(Y_1 = 0) = 0$. Then

$$P_{s'_\delta}(Y_{B(\delta(w))} = 0) = \sum_{\substack{t=(t_1, \dots, t_N)' \in \{0, 1\}^N, \\ y=(y_{t_1}, \dots, y_{t_N})' \in \{0, 1\}^N \\ y_{t_j}=1 \text{ if } t_j=1 \text{ for } j=1, \dots, N}} p^{\bar{t}}(1-p)^{N-\bar{t}} \varepsilon^{N_{01y}} (1-\varepsilon)^{N-\bar{t}-N_{01y}} (1-\delta((t, y)))^\varepsilon \quad (\text{A.17})$$

exceeds zero, because $1 - \delta((0^{N'}, 1^{N'}))' = 1$ and the vectors $t = 0^N$ and $y = 1^N$ appear in the sum in (A.17). Thus, $R(\delta, s'_\delta) = 1$.

Finally, the case ($\alpha = 1$ and $r = 0$) is handled exactly as in part (i).

Part (iii). We first show the following lemma

LEMMA 1. *Let $(\alpha \in (0, 1)$ and $r \in [0, 1])$ or $(\alpha = 0$ and $r = 1)$. If $\delta \neq 0$, then there exists an $s_\delta \in \mathbb{S}$ such that $R(\delta, s_\delta) = q_\alpha(Y_0)$.*

A.2. Proof of Lemma 1

The case $q_\alpha(Y_0) = 0$ is trivial. Simply take s_δ such that $P_{s_\delta}(Y_0 = 0) = P_{s_\delta}(Y_1 = 0) = 1$. Thus, from now on, we assume $q_\alpha(Y_0) > 0$.

Assume first $\alpha \in (0, 1)$. Because $\delta \neq 0$, there exists a vector

$$\tilde{w} = (y_{1,1}, \dots, y_{1,N}) \text{ such that } \delta(\tilde{w}) > 0, \quad (\text{A.18})$$

where $y_{1,i} \in [0, 1]$ for $i = 1, \dots, N$. (When $N = 0$, then we don't need to define $\tilde{w}$).

Definition of Distributions of Y_0 and Y_1 : We now define an $s_\delta \in \mathbb{S}$ for which $R(\delta, s_\delta) = q_\alpha(Y_0)$ and under $s_\delta \in \mathbb{S}$, Y_0 and Y_1 have independent discrete marginals. Namely, Y_0 is defined by $P_{s_\delta}(Y_0 = 0) = \alpha - \varepsilon$ for some small $\varepsilon > 0$ (and obviously $\varepsilon < \alpha$) to be further restricted below and $P_{s_\delta}(Y_0 = q_\alpha(Y_0)) = 1 - (\alpha - \varepsilon)$. Thus, $q_{s_\delta, \alpha}(Y_0) = q_\alpha(Y_0)$.

When $N = 0$, set $P_{s_\delta}(Y_1 = 0) = 1$ and note that $P_{s_\delta}(Y_{B(\delta)} = 0) = \delta + (1 - \delta)(\alpha - \varepsilon)$ which exceeds α for ε small enough (using $\alpha < 1$). Thus, $q_{s_\delta, \alpha}(Y_1) = 0$ and $q_{s_\delta, \alpha}(Y_{B(\delta)}) = 0$ which implies the desired result $R(\delta, s_\delta) = q_\alpha(Y_0)$.

When $N \geq 1$, the discrete distribution of Y_1 under s_δ is defined by

$$P_{s_\delta}(Y_1 = 0) = \frac{\alpha + 1}{2} + \frac{1}{N} \left[1 - \frac{\alpha + 1}{2} \right] \sum_{j=1}^N I(y_{1,j} = 0),$$

$$P_{s_\delta}(Y_1 = y_{1,i}) = \frac{1}{N} \left[1 - \frac{\alpha + 1}{2} \right] \sum_{j=1}^N I(y_{1,j} = y_{1,i}) \text{ for all } i = 1, \dots, N \text{ s.t. } y_{1,i} \neq 0, \quad (\text{A.19})$$

where the $y_{1,i}$ for $i = 1, \dots, N$ appear as the components of $\tilde{w}$ in (A.18). Note that all probabilities in (A.19) are properly defined, that is, they are all in the interval $(0, 1]$, and sum up to 1. Just for clarity, $\sum_{j=1}^N I(y_{1,j} = 0)$ equals the number of zero components in the vector $\tilde{w}$. Note that several of the $y_{1,i}$ for $i = 1, \dots, N$ may be identical and therefore Y_1 may be supported on strictly fewer than $N + 1$ points.

With these definitions, we obtain $q_{s_\delta, \alpha}(Y_1) = 0$ (because $0.5(\alpha + 1) > \alpha$) and

$$\begin{aligned}
 & P_{s_\delta}(Y_{B(\delta(w))} = 0) \\
 &= \sum_{\substack{\widehat{w}=(w_1, \dots, w_N) \\ w_i \in \{0, y_{1,1}, \dots, y_{1,N}\}, i=1, \dots, N}} \prod_{i=1}^N P_{s_\delta}(Y_1 = w_i) \\
 & \quad [\delta(\widehat{w})P_{s_\delta}(Y_1 = 0) + (1 - \delta(\widehat{w}))P_{s_\delta}(Y_0 = 0)] \\
 &= \left\{ \sum_{\substack{\widehat{w}=(w_1, \dots, w_N) \\ w_i \in \{0, y_{1,1}, \dots, y_{1,N}\}, i=1, \dots, N}} \prod_{i=1}^N P_{s_\delta}(Y_1 = w_i) \delta(\widehat{w}) [P_{s_\delta}(Y_1 = 0) - P_{s_\delta}(Y_0 = 0)] \right\} \\
 & \quad + P_{s_\delta}(Y_0 = 0) \\
 &\geq \left\{ \prod_{i=1}^N P_{s_\delta}(Y_1 = y_{1,i}) \delta(\widetilde{w}) \left[\frac{\alpha + 1}{2} - (\alpha - \varepsilon) \right] \right\} + \alpha - \varepsilon, \tag{A.20}
 \end{aligned}$$

where the inequality is obtained by setting $\delta(\widehat{w}) = 0$ except for when $\widehat{w} = \widetilde{w}$, using that $P_{s_\delta}(Y_1 = 0) - P_{s_\delta}(Y_0 = 0) \geq \frac{\alpha+1}{2} - (\alpha - \varepsilon) > 0$. We claim that $P_{s_\delta}(Y_{B(\delta(w))} = 0) > \alpha$ (and thus $q_{s_\delta, \alpha}(Y_{B(\delta(w))}) = 0$) for sufficiently small $\varepsilon > 0$. But the claim is equivalent to

$$\prod_{i=1}^N P_{s_\delta}(Y_1 = y_{1,i}) \delta(\widetilde{w}) \left[\frac{1 - \alpha}{2} + \varepsilon \right] > \varepsilon, \tag{A.21}$$

which clearly holds true for $\varepsilon > 0$ sufficiently small (using $\alpha < 1$).

When $\alpha = 0$ (in which case, we take $r = 1$), we can use the same proof structure except we use s_δ such that $P_{s_\delta}(Y_0 = q_\alpha(Y_0)) = 1$ (which implies $q_{s_\delta, \alpha}(Y_0) = q_\alpha(Y_0)$) and the distribution for Y_1 defined in (A.19) with $\alpha = 0$. $\square$

Note throughout that for $s \in \mathbb{S}$, we have $q_\alpha(Y_0) = q_{s, \alpha}(Y_0)$.

Consider first the case $q_\alpha(Y_0) > 1/2$. For $\delta^0 \equiv 0$, we will show next that

$$\max_{s \in \mathbb{S}} R(\delta^0, s) = 1 - q_\alpha(Y_0). \tag{A.22}$$

The statement in (A.22) follows easily because for any $s \in \mathbb{S}$ for which $q_{s, \alpha}(Y_1) = 1$, we obtain $R(\delta^0, s) = 1 - q_\alpha(Y_0)$. Furthermore, because always

$$q_{s, \alpha}(Y_{B(\delta(w))}) \in [\min\{q_{s, \alpha}(Y_0), q_{s, \alpha}(Y_1)\}, \max\{q_{s, \alpha}(Y_0), q_{s, \alpha}(Y_1)\}] \tag{A.23}$$

(as proven in Lemma 3(i) below) $1 - q_{s, \alpha}(Y_0)$ is the maximal possible regret for any $s \in \mathbb{S}$ for which $q_{s, \alpha}(Y_1) > q_{s, \alpha}(Y_0)$. For any $s \in \mathbb{S}$, such that $q_{s, \alpha}(Y_1) \leq q_{s, \alpha}(Y_0)$ regret is zero. That proves (A.22).

Because when $q_\alpha(Y_0) > 1/2$, we have $1 - q_{s, \alpha}(Y_0) < q_{s, \alpha}(Y_0)$, (A.22) combined with the lemma implies the desired result when $q_\alpha(Y_0) > 1/2$.

Next, consider the case $q_\alpha(Y_0) = 1/2$. Statement (A.22) (that also holds when $q_\alpha(Y_0) = 1/2$), Lemma 1, together with the fact that regret is bounded by $1/2$ (by (A.23) when $q_{s, \alpha}(Y_0) = 1/2$), establish that all treatment rules are minimax regret.

Finally, consider the case $q_\alpha(Y_0) < 1/2$. We have $\max_{s \in \mathbb{S}} R(\delta^*, s) = q_{s, \alpha}(Y_0)$ for $\delta^* = 1$. Lemma 1 establishes that for $\delta \neq 0$ $\max_{s \in \mathbb{S}} R(\delta, s) \geq q_{s, \alpha}(Y_0)$ while for $\delta^0 \equiv 0$, $\max_{s \in \mathbb{S}} R(\delta^0, s) = 1 - q_{s, \alpha}(Y_0) > q_{s, \alpha}(Y_0)$. It follows that $\delta^* = 1$ is a minimax rule.

Finally, the case where $\alpha = 1$ and $r = 0$ is proven as in part (i). $\square$

A.3. Proof of Corollary 1

(a) This statement follows as a corollary from the proof of Proposition 1 because that proof only used Bernoulli distributions for nature for parts (i) and (ii) and discrete distributions in part (iii).

(b) (i) Consider an arbitrary treatment rule δ . Consider a state of nature $s = s_{\alpha, \varepsilon, n}$ for $n \in \mathbb{N}$ such that Y_0 has density n on $[0, 1/n]$ and Y_1 has density $f = f_{\alpha, \varepsilon, n}$ which equals $n(\alpha - \varepsilon)$ on $[0, 1/n]$ and $n(1 - \alpha + \varepsilon)$ on $[1 - 1/n, 1]$ (and zero elsewhere) for some small $\varepsilon \in [0, \alpha/2]$ to be specified more precisely below. Note that these continuous distributions are chosen in order to approximate as $n \rightarrow \infty$ the Bernoulli distributions used in the proof of Proposition 1(i). With these definitions,

$$q_{s, \alpha}(Y_0) = \alpha/n \text{ and } q_{s, \alpha}(Y_1) = 1 - 1/n + \varepsilon/(n(1 - \alpha + \varepsilon)). \quad (\text{A.24})$$

Consider an arbitrary $q \in (0, 1)$. Then for all n such that $1/n < q < 1 - 1/n$

$$\begin{aligned} P_s(Y_{B(\delta(w))} \leq q) &= \int_0^{1/n} n \dots \int_0^{1/n} n \int_0^1 f(y_{11}) \dots \int_0^1 f(y_{1N_1}) \\ &\quad [\delta(y)(\alpha - \varepsilon) + (1 - \delta(y))] dy_{1N_1} \dots dy_{11} dy_{0N_0} \dots dy_{01} \\ &\geq (\alpha - \varepsilon) \int_0^{1/n} n \dots \int_0^{1/n} n \int_0^1 f(y_{11}) \dots \int_0^1 f(y_{1N_1}) dy_{1N_1} \dots dy_{11} dy_{0N_0} \dots dy_{01} \\ &= (\alpha - \varepsilon), \end{aligned} \quad (\text{A.25})$$

where $y := (y_{01}, \dots, y_{0N_0}, y_{11}, \dots, y_{1N_1})'$ and the inequality comes from picking $\delta^1 \equiv 1$ given that $\alpha - \varepsilon - 1 < 0$.

Note that if $P_s(\delta(w) = 1) < 1$ for one choice of $\varepsilon \in [0, \alpha/2]$, then it holds for any $\varepsilon \in [0, \alpha/2]$.

Case 1: $P_s(\delta(w) = 1) < 1$. By the same steps as in (A.25) with $\varepsilon = 0$, we obtain $P_s(Y_{B(\delta(w))} \leq q) > \alpha$ and, by continuity in ε , then also for $\varepsilon > 0$ sufficiently small. It follows that the α -quantile of $Y_{B(\delta(w))}$ is not greater than q .

Case 2: $P_s(\delta(w) = 1) = 1$. In that case, consider a state of nature $\tilde{s} = \tilde{s}_{\alpha, \varepsilon, n}$ for $n \in \mathbb{N}$ such that Y_1 has density n on $[0, 1/n]$ and Y_0 has density $\tilde{f} = \tilde{f}_{\alpha, \varepsilon, n}$ which equals $n(\alpha - \varepsilon)$ on $[0, 1/n]$ and $n(1 - \alpha + \varepsilon)$ on $[1 - 1/n, 1]$ (and zero elsewhere). Thus,

$$q_{\tilde{s}, \alpha}(Y_1) = \alpha/n \text{ and } q_{\tilde{s}, \alpha}(Y_0) = 1 - 1/n + \varepsilon/(n(1 - \alpha + \varepsilon)) \quad (\text{A.26})$$

and, analogously to (A.25),

$$\begin{aligned} P_{\tilde{s}}(Y_{B(\delta(w))} \leq q) &= \int_0^{1/n} \tilde{f}(y_{01}) \dots \int_0^{1/n} \tilde{f}(y_{0N_0}) \int_0^{1/n} n \dots \int_0^{1/n} n [\delta(y) + (1 - \delta(y))(\alpha - \varepsilon)] \\ &\quad dy_{1N_1} \dots dy_{11} dy_{0N_0} \dots dy_{01} \\ &= \int_0^{1/n} \tilde{f}(y_{01}) \dots \int_0^{1/n} \tilde{f}(y_{0N_0}) \int_0^{1/n} n \dots \int_0^{1/n} n dy_{1N_1} \dots dy_{11} dy_{0N_0} \dots dy_{01} \\ &\quad + \int \dots \int_{A_{\delta^1}} \tilde{f}(y_{01}) \dots \tilde{f}(y_{0N_0}) \int_0^{1/n} n \dots \int_0^{1/n} n [\delta(y) + (1 - \delta(y))(\alpha - \varepsilon)] \\ &\quad dy_{1N_1} \dots dy_{11} dy_{0N_0} \dots dy_{01} \\ &\geq (\alpha - \varepsilon)^{N_0} + (\alpha - \varepsilon) \int \dots \int_{A_{\delta^1}} \tilde{f}(y_{01}) \dots \tilde{f}(y_{0N_0}) dy_{0N_0} \dots dy_{01} \\ &= (\alpha - \varepsilon)^{N_0} + (\alpha - \varepsilon)(1 - (\alpha - \varepsilon)^{N_0}), \end{aligned} \quad (\text{A.27})$$

where $A_n := [0, 1]^{N_0} \setminus [0, 1/n]^{N_0}$, the second equality uses that $\delta(w) = 1$ a.s. on $[0, 1/n]^N$ (under s and therefore also under $\tilde{s}$), and the inequality is obtained by taking $\delta(y) = 0$ on the domain of integration. If ε was equal to zero, the final expression in (A.27) would be strictly larger than α . By continuity in ε , that is then also true for sufficiently small $\varepsilon > 0$. It follows that the α -quantile of $Y_{B(\delta(w))}$ is not greater than q .

As q can be chosen arbitrarily small in that construction it follows that, both in Cases 1 and 2, regret arbitrarily close to 1 can be inflicted by nature given (A.24) and (A.26).

Parts (ii) and (iii) follow from similar constructions. $\square$

A.4. Proof of the Statements in Comment 3 below Corollary 1

Note first that $q(\delta)$ indeed exists because cdfs are continuous from the left. For example, $q(0) = q_\alpha(Y_0)$ and $q(1) = 0$. Also note that for any $q \in [0, 1]$, the function $g(\delta) := \delta + (1 - \delta)F_{Y_0}(q)$ is weakly increasing in δ . It follows that $q(\delta)$ is weakly decreasing in δ . Therefore, $q_\alpha(Y_0) - q(\delta)$ is weakly increasing in δ .

If $\delta < 1$, then regret of $1 - q_\alpha(Y_0)$ can be achieved by nature by defining the distribution of Y_1 as $P_s(Y_1 = 0) = \alpha - \varepsilon$ and $P_s(Y_1 = 1) = 1 - (\alpha - \varepsilon)$ for some small $\varepsilon > 0$ (to be specified further below). Namely, with that definition, for $q_\alpha(Y_0) < 1$, we have

$$P_s(Y_{B(\delta)} \leq q_\alpha(Y_0)) = \delta(\alpha - \varepsilon) + (1 - \delta)F_{Y_0}(q_\alpha(Y_0)), \quad (\text{A.28})$$

which exceeds α for ε small enough because $F_{Y_0}(q_\alpha(Y_0)) > \alpha$ and $\delta < 1$. And of course $P_s(Y_{B(\delta)} \leq q_\alpha(Y_0)) \geq \alpha$ when $q_\alpha(Y_0) = 1$. Because the α -quantile of $Y_{B(\delta)}$ can clearly not be smaller than $q_\alpha(Y_0)$ it follows that it is equal to $q_\alpha(Y_0)$.

But then, in the case $q_\alpha(Y_0) < 1/2$, it follows that for any $\delta \in [0, 1]$, $\max_{s \in \mathbb{S}} R(\delta, s) \geq 1 - q_\alpha(Y_0) > q_\alpha(Y_0)$. Given that for $\delta^1 \equiv 1$ maximal regret equals $q_\alpha(Y_0)$, it is the unique minimax regret rule. When $q_\alpha(Y_0) = 1/2$, regret cannot exceed $1/2$ and given regret of $1 - q_\alpha(Y_0)$ can always be achieved by the argument above (for $\delta < 1$ and for $q_\alpha(Y_0) = 1/2$ also for $\delta = 1$), it follows that all rules are minimax regret.

Furthermore, for any $\delta \in [0, 1]$, regret of $q_\alpha(Y_0) - q(\delta)$ can be achieved by nature by defining s , the distribution of Y_1 , by $P_s(Y_1 = 0) = 1$. Then $P_s(Y_{B(\delta)} \leq q(\delta)) = \delta + (1 - \delta)F_{Y_0}(q(\delta))$ which by (2.23) and $r = 0$ implies $q_{s,\alpha}(Y_{B(\delta)}) = q(\delta)$. By construction, no bigger regret than $q_\alpha(Y_0) - q(\delta)$ can be achieved by nature for setups where $q_\alpha(Y_1) \leq q_\alpha(Y_0)$. That proves the claim in the case where $q_\alpha(Y_0) > 1/2$. $\square$

A.5. Proof of Corollary 2

Part (i). Let δ be an arbitrary treatment rule. We will construct a state of nature s_δ that yields regret equal to 1. Consider the case where all marginals under s_δ are independent Bernoulli distributions and define

$$a_{0x_k} = P_{s_\delta}(Y_{0,x_k} = 0) \text{ and } a_{1x_k} = P_{s_\delta}(Y_{1,x_k} = 0) \quad (\text{A.29})$$

for $k = 1, \dots, K$ (and, of course, $1 - a_{0x_k} = P_{s_\delta}(Y_{0,x_k} = 1)$ and $1 - a_{1x_k} = P_{s_\delta}(Y_{1,x_k} = 1)$). Notationwise, we index probabilities by those distributions that they depend on. Then,

$$P_{s_\delta, F_X}(Y_{B(\delta_X(w))}, X = 0) = \sum_{k=1}^K P_{F_X}(X = x_k) P_{s_\delta, F_X}(Y_{B(\delta_{x_k}(w))}, x_k = 0) \quad (\text{A.30})$$

and

$$\begin{aligned}
 & P_{s_\delta, F_X}(Y_{B(\delta_{x_k}(w))}, x_k = 0) \\
 &= \sum_{\substack{\tilde{x}=(\tilde{x}_1, \dots, \tilde{x}_N) \in \mathcal{X}^N \\ y=(y_{t_1}, \dots, y_{t_N}) \in \{0, 1\}^N}} \prod_{l=1}^{N_0} P_{F_X}(X = \tilde{x}_l) a_{0\tilde{x}_l}^{1-y_{t_l}} (1 - a_{0\tilde{x}_l})^{y_{t_l}} \\
 &\quad \prod_{l'=N_0+1}^N P_{F_X}(X = \tilde{x}_{l'}) a_{1\tilde{x}_{l'}}^{1-y_{t_{l'}}} (1 - a_{1\tilde{x}_{l'}})^{y_{t_{l'}}} \\
 &\times [\delta_{x_k}(\tilde{w}(\tilde{x}, y)) a_{1x_k} + (1 - \delta_{x_k}(\tilde{w}(\tilde{x}, y))) a_{0x_k}], \tag{A.31}
 \end{aligned}$$

where in the last line $\tilde{w}(\tilde{x}, y)$ denotes the sample associated with $\tilde{x}$ and y . Note that expressions like $P_{s_\delta, F_X}(Y_{B(\delta_{x_k}(w))}, x_k = 0)$ are still indexed by F_X because the distribution of w depends on F_X . In the case $(a_{0x_k}, a_{1x_k}) = (1, \alpha)$ for all $k = 1, \dots, K$, (A.31) becomes

$$\begin{aligned}
 & P_{s_\delta, F_X}(Y_{B(\delta_{x_k}(w))}, x_k = 0) = \sum_{\substack{\tilde{x}=(\tilde{x}_1, \dots, \tilde{x}_N) \in \mathcal{X}^N \\ y'_{N_0+1}, \dots, y'_N \in \{0, 1\}^{N_1}}} \prod_{l=1}^{N_0} P_{F_X}(X = \tilde{x}_l) \\
 &\quad \prod_{l'=N_0+1}^N P_{F_X}(X = \tilde{x}_{l'}) \alpha^{1-y_{l'}} (1 - \alpha)^{y_{l'}} \times [(\alpha - 1) \delta_{x_k}(\tilde{w}(\tilde{x}, y)) + 1] \\
 &\geq [\sum_{\substack{\tilde{x}=(\tilde{x}_1, \dots, \tilde{x}_N) \in \mathcal{X}^N \\ (y'_{N_0+1}, \dots, y'_N) \in \{0, 1\}^{N-N_0}}} \prod_{l=1}^{N_0} P_{F_X}(X = \tilde{x}_l) \prod_{l'=N_0+1}^N P_{F_X}(X = \tilde{x}_{l'}) \alpha^{1-y_{l'}} (1 - \alpha)^{y_{l'}}] \alpha \\
 &= \alpha, \tag{A.32}
 \end{aligned}$$

where the inequality follows from taking $\delta_{x_k}(\tilde{w}(\tilde{x}, y)) = 1$.

If for some $k \in \{1, \dots, K\}$ $\delta_{x_k}(\tilde{w}(\tilde{x}, y)) < 1$ for some $\tilde{w}(\tilde{x}, y)$ (with $\tilde{x} = (\tilde{x}_1, \dots, \tilde{x}_N) \in \mathcal{X}^N$ and $y = (0, \dots, 0, y_{t_{N_0+1}}, \dots, y_{t_N})$ with $y_{t_j} \in \{0, 1\}$ for $j = N_0 + 1, \dots, N$) it follows that $P_{s_\delta, F_X}(Y_{B(\delta_{x_k}(w))}, x_k = 0) > \alpha$ and by (A.30) therefore also $P_{s_\delta, F_X}(Y_{B(\delta_X(w))}, X = 0) > \alpha$. Notice that $P_{s_\delta, F_X}(Y_{B(\delta_X(w))}, X = 0)$ is continuous in a_{1x_k} for $k = 1, \dots, K$ and therefore $P_{s_\delta, F_X}(Y_{B(\delta_X(w))}, X = 0) > \alpha$ when instead of $(a_{0x_k}, a_{1x_k}) = (1, \alpha)$ for $k = 1, \dots, K$, s_δ is defined by $(a_{0x_k}, a_{1x_k}) = (1, \alpha - \varepsilon)$ for $k = 1, \dots, K$ for some small enough $\varepsilon > 0$. But then for that choice of s_δ , regret equals 1 (as can be seen by comparing to the quantile obtained for the treatment rule $\delta \equiv 1$) as desired.

If instead for all $k \in \{1, \dots, K\}$ $\delta_{x_k}(\tilde{w}(\tilde{x}, y)) = 1$ for all $\tilde{w}(\tilde{x}, y)$ (with $\tilde{x} = (\tilde{x}_1, \dots, \tilde{x}_N) \in \mathcal{X}^N$ and $y = (0, \dots, 0, y_{t_{N_0+1}}, \dots, y_{t_N})$ with $y_{t_j} \in \{0, 1\}$ for $j = N_0 + 1, \dots, N$), nature can cause regret of 1 by using distributions defined by $(a_{0x_k}, a_{1x_k}) = (\alpha - \varepsilon, 1)$ for some small enough $\varepsilon > 0$. Namely, note first that for s_δ defined by $(a_{0x_k}, a_{1x_k}) = (\alpha, 1)$ for $k = 1, \dots, K$, we obtain from (A.31)

$$\begin{aligned}
 & P_{s_\delta, F_X}(Y_{B(\delta_{x_k}(w))}, x_k = 0) \\
 &= \sum_{\substack{\tilde{x}=(\tilde{x}_1, \dots, \tilde{x}_N) \in \mathcal{X}^N \\ y=(y_{t_1}, \dots, y_{t_{N_0}}, 0, \dots, 0) \in \{0, 1\}^{N_0} \times \{0\}^{N_1}}} \prod_{l=1}^{N_0} P_{F_X}(X = \tilde{x}_l) \alpha^{1-y_{t_l}} (1 - \alpha)^{y_{t_l}} \\
 &\quad \prod_{l'=N_0+1}^N P_{F_X}(X = \tilde{x}_{l'}) \\
 &\times [\delta_{x_k}(\tilde{w}(\tilde{x}, y)) (1 - \alpha) + \alpha] \\
 &\geq \alpha^{N_0} \sum_{\tilde{x}=(\tilde{x}_1, \dots, \tilde{x}_N) \in \mathcal{X}^N} \prod_{l=1}^{N_0} P_{F_X}(X = \tilde{x}_l) \\
 &+ \alpha \sum_{\substack{\tilde{x}=(\tilde{x}_1, \dots, \tilde{x}_N) \in \mathcal{X}^N \\ y=(y_{t_1}, \dots, y_{t_{N_0}}, 0, \dots, 0) \in \{0, 1\}^{N_0} \times \{0\}^{N_1}, y \neq 0^N}} \prod_{l=1}^{N_0} P_{F_X}(X = \tilde{x}_l) \alpha^{1-y_{t_l}} (1 - \alpha)^{y_{t_l}}
 \end{aligned}$$

$$\begin{aligned}
 & (\prod_{l'=N_0+1}^N P_{F_X}(X = \tilde{x}_{l'})) \\
 &= \alpha^{N_0} + \alpha - \alpha [\sum_{\tilde{x}=(\tilde{x}_1, \dots, \tilde{x}_N) \in \mathcal{X}^N} \prod_{l=1}^{N_0} (P_{F_X}(X = \tilde{x}_l) \alpha) (\prod_{l'=N_0+1}^N P_{F_X}(X = \tilde{x}_{l'}))] \\
 &= \alpha^{N_0} + \alpha - \alpha^{N_0+1},
 \end{aligned} \tag{A.33}$$

where the inequality follows from setting $\delta_{x_k}(\tilde{w}(\tilde{x}, y)) = 0$ for all elements in the sum except for those when $y = 0^N$ in which case we use $\delta_{x_k}(\tilde{w}(\tilde{x}, y)) = 1$. The second to last equality follows by first considering a sum over all $y = (y_{t_1}, \dots, y_{t_{N_0}}, 0, \dots, 0) \in \{0, 1\}^{N_0} \times \{0\}^{N_1}$ in the second sum and then subtracting the summand associated with $y = 0^N$. The final expression is strictly larger than α which by an argument as above allows us to conclude that $P_{s_\delta, F_X}(Y_{B(\delta_{x_k}(w)), x_k} = 0) > \alpha$ when s_δ is defined by $(a_{0x_k}, a_{1x_k}) = (\alpha - \varepsilon, 1)$ for $k = 1, \dots, K$ for some small enough $\varepsilon > 0$.

The proof of part (ii) is very similar to part (i) and omitted.

Part (iii). The key input of the proof is the following lemma.

LEMMA 2. *For any treatment rule $\delta \in \mathbb{D}$ such that $\delta \neq 0$, there exists an $s_\delta \in \mathbb{S}$ such that $R(\delta, s_\delta) = q_\alpha(Y_{0,X})$.*

A.6. Proof of Lemma 2

We define an $s_\delta \in \mathbb{S}$ by letting all marginals (Y_{t, x_k}) be independent and discrete, namely, for an $\varepsilon > 0$ (to be specified later), let $P_{s_\delta}(Y_{0, x_k} = 0) = \alpha - \varepsilon$ and $P_{s_\delta}(Y_{0, x_k} = q_\alpha(Y_{0, X})) = 1 - (\alpha - \varepsilon)$ for every $x_k \in \mathcal{X}$. Because $P_{s_\delta, F_X}(Y_{0, X} = 0) = \sum_{k=1}^K P_{F_X}(X = x_k) P_{s_\delta}(Y_{0, x_k} = 0) = \alpha - \varepsilon$, it follows that $q_{s_\delta, F_X, \alpha}(Y_{0, X}) = q_\alpha(Y_{0, X})$ as required.

For $N = 0$, complete the definition of s_δ by setting $P_{s_\delta}(Y_{1, x_k} = 0) = 1$ for every $x_k \in \mathcal{X}$. Wlog assume $\delta_{x_l} > 0$. Note that $P_{s_\delta}(Y_{B(\delta_{x_l}), x_l} = 0) = \delta_{x_l} + (1 - \delta_{x_l})(\alpha - \varepsilon)$ which, given that $\delta_{x_l} > 0$, is strictly larger than α when $\varepsilon > 0$ is chosen sufficiently small. For every $x_k \in \mathcal{X}$ such that $x_k \neq x_l$, we have $P_{s_\delta}(Y_{B(\delta_{x_k}), x_k} = 0) \geq \alpha - \varepsilon$. Therefore,

$$\begin{aligned}
 & P_{s_\delta, F_X}(Y_{B(\delta_X), X} = 0) \\
 &= \sum_{k=1}^K P_{F_X}(X = x_k) P_{s_\delta}(Y_{B(\delta_{x_k}), x_k} = 0) \\
 &\geq P_{F_X}(X = x_l)(\delta_{x_l} + (1 - \delta_{x_l})(\alpha - \varepsilon)) + \sum_{k=1, k \neq l}^K P_{F_X}(X = x_k)(\alpha - \varepsilon) \\
 &= P_{F_X}(X = x_l)\delta_{x_l}(1 - (\alpha - \varepsilon)) + (\alpha - \varepsilon) \\
 &> \alpha,
 \end{aligned} \tag{A.34}$$

where the second equality follows by including the summand $P_{F_X}(X = x_l)(\alpha - \varepsilon)$ into the summation sign, and the inequality holds for $\varepsilon > 0$ chosen small enough.

Next consider the case $N > 0$. Because $\delta \neq 0$, there exists some vector $\tilde{w} = ((\tilde{x}_1, \tilde{y}_1), \dots, (\tilde{x}_N, \tilde{y}_N))$ with $(\tilde{x}_i, \tilde{y}_i) \in \mathcal{X} \times [0, 1]$ for every $i = 1, \dots, N$ and some covariate $x_l \in \mathcal{X}$ such that $\delta_{x_l}(\tilde{w}) > 0$. Similarly to the proof of Proposition 1(iii), for every $x_k \in \mathcal{X}$,

define the distribution of Y_{1,x_k} under s_δ as

$$\begin{aligned} P_{s_\delta}(Y_{1,x_k} = 0) &= \frac{\alpha + 1}{2} + \frac{1}{N} \left[1 - \frac{\alpha + 1}{2} \right] \sum_{j=1}^N I(\tilde{y}_j = 0), \\ P_{s_\delta}(Y_{1,x_k} = \tilde{y}_i) &= \frac{1}{N} \left[1 - \frac{\alpha + 1}{2} \right] \sum_{j=1}^N I(\tilde{y}_j = \tilde{y}_i) \text{ for } i = 1, \dots, N \text{ such that } \tilde{y}_i \neq 0. \end{aligned} \quad (\text{A.35})$$

Because $P_{s_\delta, F_X}(Y_{1,X} = 0) = \sum_{k=1}^K P_{F_X}(X = x_k) P_{s_\delta}(Y_{1,x_k} = 0)$, it then follows that $q_{s, F_X, \alpha}(Y_{1,X}) = 0$. Now, focus on $P_{s_\delta, F_X}(Y_{B(\delta_{x_l}(w)), x_l} = 0)$. Furthermore,

$$\begin{aligned} &P_{s_\delta, F_X}(Y_{B(\delta_{x_l}(w)), x_l} = 0) \\ &= \sum_{\substack{w=((a_1, b_1), \dots, (a_N, b_N)); \\ a_i \in \mathcal{X}; \ b_i \in \{0, \tilde{y}_1, \dots, \tilde{y}_N\}, \\ i=1, \dots, N}} \prod_{i=1}^N P_{s_\delta, F_X}(Y_{1, a_i} = b_i, X = a_i) \\ &\quad (\delta_{x_l}(w) P_{s_\delta}(Y_{1, x_l} = 0) + (1 - \delta_{x_l}(w)) P_{s_\delta}(Y_{0, x_l} = 0)) \\ &= P_{s_\delta}(Y_{0, x_l} = 0) + \sum_{\substack{w=((a_1, b_1), \dots, (a_N, b_N)); \\ a_i \in \mathcal{X}; \ b_i \in \{0, \tilde{y}_1, \dots, \tilde{y}_N\}, \\ i=1, \dots, N}} \prod_{i=1}^N P_{s_\delta, F_X}(Y_{1, a_i} = b_i, X = a_i) (\delta_{x_l}(w) \\ &\quad (P_{s_\delta}(Y_{1, x_l} = 0) - P_{s_\delta}(Y_{0, x_l} = 0)) \\ &\geq \alpha - \varepsilon + \prod_{i=1}^N P_{s_\delta}(Y_{1, \tilde{y}_i} = \tilde{y}_i) P_{F_X}(X = \tilde{x}_i) \delta_{x_l}(\tilde{w}) \left(\frac{\alpha + 1}{2} - (\alpha - \varepsilon) \right) \\ &> \alpha, \end{aligned} \quad (\text{A.36})$$

where the first inequality follows by replacing $\delta_{x_l}(w)$ by zero for every $w \neq \tilde{w}$ and the last inequality holds for small enough $\varepsilon > 0$. By the same derivations, for any other $x_k \neq x_l$, it follows that $P_{s_\delta, F_X}(Y_{B(\delta_{x_k}(w)), x_k} = 0) \geq \alpha - \varepsilon$. Therefore, for sufficiently small $\varepsilon > 0$, we have

$$\begin{aligned} &P_{s_\delta, F_X}(Y_{B(\delta_X(w)), X} = 0) \\ &= \sum_{k=1}^K P_{F_X}(X = x_k) P_{s_\delta}(Y_{B(\delta_{x_k}(w)), x_k} = 0) \\ &\geq \alpha - \varepsilon + P_{F_X}(X = x_l) \prod_{i=1}^N P_{s_\delta}(Y_{1, \tilde{y}_i} = \tilde{y}_i) P_{F_X}(X = \tilde{x}_i) \delta_{x_l}(\tilde{w}) \left(\frac{\alpha + 1}{2} - (\alpha - \varepsilon) \right) \\ &> \alpha, \end{aligned} \quad (\text{A.37})$$

where for the inequality we use $P_{s_\delta}(Y_{B(\delta_{x_k}(w)), x_k} = 0) \geq \alpha - \varepsilon$ and the second to last line in (A.36). Therefore, we have $q_{s, F_X, \alpha}(Y_{0,X}) = q_\alpha(Y_{0,X})$ and $q_{s, F_X, \alpha}(Y_{1,X}) = q_{s, F_X, \alpha}(Y_{B(\delta_X(w)), X}) = 0$. Hence, the statement of the lemma follows. $\square$

Given Lemma 2, the remainder of the proof of part (iii) is exactly the same as the last part of the proof of Proposition 1(iii). The only detail to think about is establishing that

$$q_{s, F_X, \alpha}(Y_{B(\delta_X(w))}, X) \in [\min\{q_{s, F_X, \alpha}(Y_{0, X}), q_{s, F_X, \alpha}(Y_{1, X})\}, \max\{q_{s, F_X, \alpha}(Y_{0, X}), q_{s, F_X, \alpha}(Y_{1, X})\}]. \quad (\text{A.38})$$

The proof of that statement is analogous to the proof of Lemma 3 and therefore omitted. $\square$

The following is a technical lemma that is used in the proof of Proposition 1 above. It states that $1(q_{s, \alpha}(Y_1) \geq q_{s, \alpha}(Y_0))$ is an infeasible optimal rule.

LEMMA 3. *For any given $r \in [0, 1]$ and arbitrary $s \in \mathbb{S}$, we have (i)*

$$q_{s, \alpha}(Y_{B(\delta(w))}) \in [\min\{q_{s, \alpha}(Y_0), q_{s, \alpha}(Y_1)\}, \max\{q_{s, \alpha}(Y_0), q_{s, \alpha}(Y_1)\}]$$

and (ii)

$$\max_{\delta \in \mathbb{D}} u(\delta, s) = \max_{\delta \in \mathbb{D}} q_{s, \alpha}(Y_{B(\delta(w))}) = \max\{q_{s, \alpha}(Y_0), q_{s, \alpha}(Y_1)\}.$$

A.7. Proof of Lemma 3

(i) Denote by $\bar{q}_{0, \alpha}$, $\bar{q}_{1, \alpha}$, and $\bar{q}_{\delta, \alpha}$ the largest α -quantile (meaning employing $r = 1$ in definition (2.5)) of Y_0 , Y_1 , and $Y_{B(\delta(w))}$, respectively. Likewise, denote by $\underline{q}_{0, \alpha}$, $\underline{q}_{1, \alpha}$, and $\underline{q}_{\delta, \alpha}$ the smallest α -quantile (meaning employing $r = 0$ in definition (2.5)) of Y_0 , Y_1 , and $Y_{B(\delta(w))}$, respectively. As in (A.1),

$$P_s(Y_{B(\delta(w))} \geq q) = P_s(B(\delta(w)) = 1)P_s(Y_1 \geq q) + P_s(B(\delta(w)) = 0)P_s(Y_0 \geq q). \quad (\text{A.39})$$

We first show that

$$\bar{q}_{\delta, \alpha} \in [\min\{\bar{q}_{0, \alpha}, \bar{q}_{1, \alpha}\}, \max\{\bar{q}_{0, \alpha}, \bar{q}_{1, \alpha}\}]. \quad (\text{A.40})$$

Indeed, if $\bar{q}_{\delta, \alpha} > \max\{\bar{q}_{0, \alpha}, \bar{q}_{1, \alpha}\}$, then, from (A.39),

$$\begin{aligned} P_s(Y_{B(\delta(w))} \geq \bar{q}_{\delta, \alpha}) \\ &= P_s(B(\delta(w)) = 1)P_s(Y_1 \geq \bar{q}_{\delta, \alpha}) + P_s(B(\delta(w)) = 0)P_s(Y_0 \geq \bar{q}_{\delta, \alpha}) \\ &< 1 - \alpha, \end{aligned} \quad (\text{A.41})$$

a contradiction to the definition of an α -quantile. On the other hand, if $\bar{q}_{\delta, \alpha} < \min\{\bar{q}_{0, \alpha}, \bar{q}_{1, \alpha}\}$, then there exists $\varepsilon > 0$ such that $\bar{q}_{\delta, \alpha} + \varepsilon < \min\{\bar{q}_{0, \alpha}, \bar{q}_{1, \alpha}\}$. But then

$$\begin{aligned} P_s(Y_{B(\delta(w))} \geq \bar{q}_{\delta, \alpha} + \varepsilon) \\ &= P_s(B(\delta(w)) = 1)P_s(Y_1 \geq \bar{q}_{\delta, \alpha} + \varepsilon) + P_s(B(\delta(w)) = 0)P_s(Y_0 \geq \bar{q}_{\delta, \alpha} + \varepsilon) \\ &\leq P_s(B(\delta(w)) = 1)P_s(Y_1 \geq \bar{q}_{1, \alpha}) + P_s(B(\delta(w)) = 0)P_s(Y_0 \geq \bar{q}_{0, \alpha}) \\ &= 1 - \alpha, \end{aligned} \quad (\text{A.42})$$

contradicting the fact that $\bar{q}_{\delta, \alpha}$ is the largest α -quantile of $Y_{B(\delta(w))}$.

By an analogous argument, it follows that

$$\underline{q}_{\delta, \alpha} \in [\min\{\underline{q}_{0, \alpha}, \underline{q}_{1, \alpha}\}, \max\{\underline{q}_{0, \alpha}, \underline{q}_{1, \alpha}\}]. \quad (\text{A.43})$$

Define the interval $I = [\min\{\underline{q}_{0,\alpha}, \underline{q}_{1,\alpha}\}, \max\{\bar{q}_{0,\alpha}, \bar{q}_{1,\alpha}\}]$.

If $I = [\underline{q}_{i,\alpha}, \bar{q}_{j,\alpha}]$ for $i, j \in \{0, 1\}$, $i \neq j$, then

$$q_{s,\alpha}(Y_{B(\delta(w))}) = r\bar{q}_{\delta,\alpha} + (1-r)\underline{q}_{\delta,\alpha} \quad (\text{A.44})$$

and thus by (A.40) and (A.43),

$$\begin{aligned} q_{s,\alpha}(Y_{B(\delta(w))}) &\leq r\bar{q}_{j,\alpha} + (1-r)\underline{q}_{j,\alpha} = q_{s,\alpha}(Y_j) \leq \max\{q_{s,\alpha}(Y_0), q_{s,\alpha}(Y_1)\} \text{ and} \\ q_{s,\alpha}(Y_{B(\delta(w))}) &\geq r\bar{q}_{i,\alpha} + (1-r)\underline{q}_{i,\alpha} = q_{s,\alpha}(Y_i) \geq \min\{q_{s,\alpha}(Y_0), q_{s,\alpha}(Y_1)\}. \end{aligned} \quad (\text{A.45})$$

For the remaining case, we have $I = [\underline{q}_{i,\alpha}, \bar{q}_{i,\alpha}]$ for an $i \in \{0, 1\}$. By (A.40) and (A.43), it follows that $[\underline{q}_{\delta,\alpha}, \bar{q}_{\delta,\alpha}] \subset I$ and $[\underline{q}_{j,\alpha}, \bar{q}_{j,\alpha}] \subset [\underline{q}_{\delta,\alpha}, \bar{q}_{\delta,\alpha}]$.

If $P_s(B(\delta(w)) = i) = 1$, then $Y_{B(\delta(w))} = Y_i$ wpl and the lemma trivially holds. If $P_s(B(\delta(w)) = i) < 1$, we must have $[\underline{q}_{j,\alpha}, \bar{q}_{j,\alpha}] = [\underline{q}_{\delta,\alpha}, \bar{q}_{\delta,\alpha}]$. If not, if, for example, $\bar{q}_{j,\alpha} < \bar{q}_{\delta,\alpha}$, then $P_s(Y_j \geq \bar{q}_{\delta,\alpha}) < 1 - \alpha$ and $P_s(Y_i \geq \bar{q}_{\delta,\alpha}) = 1 - \alpha$. But then, by (A.39), we get the contradiction

$$\begin{aligned} P_s(Y_{B(\delta(w))} \geq \bar{q}_{\delta,\alpha}) &= P_s(B(\delta(w)) = 1)P_s(Y_1 \geq \bar{q}_{\delta,\alpha}) \\ &\quad + P_s(B(\delta(w)) = 0)P_s(Y_0 \geq \bar{q}_{\delta,\alpha}) < 1 - \alpha, \end{aligned} \quad (\text{A.46})$$

where the last inequality uses $P_s(B(\delta(w)) = i) < 1$. But if $[\underline{q}_{j,\alpha}, \bar{q}_{j,\alpha}] = [\underline{q}_{\delta,\alpha}, \bar{q}_{\delta,\alpha}]$, then $q_{s,\alpha}(Y_{B(\delta(w))}) = q_{s,\alpha}(Y_j)$ and the claim in the lemma holds.

(ii) Considering the two rules $\delta^1 \equiv 1$ or $\delta^0 \equiv 0$ (i.e., the rules that pick 1 (or 0) with probability 1), it follows that $\max_{\delta \in \mathbb{D}} q_{s,\alpha}(Y_{B(\delta(w))}) \geq \max\{q_{s,\alpha}(Y_0), q_{s,\alpha}(Y_1)\}$. But given that $q_{s,\alpha}(Y_{B(\delta(w))}) \leq \max\{q_{s,\alpha}(Y_0), q_{s,\alpha}(Y_1)\}$ for any $\delta \in \mathbb{D}$ by part (i), the claim follows. $\square$

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